

3

Structure and Stereochemistry of Alkanes

Goals for Chapter 3

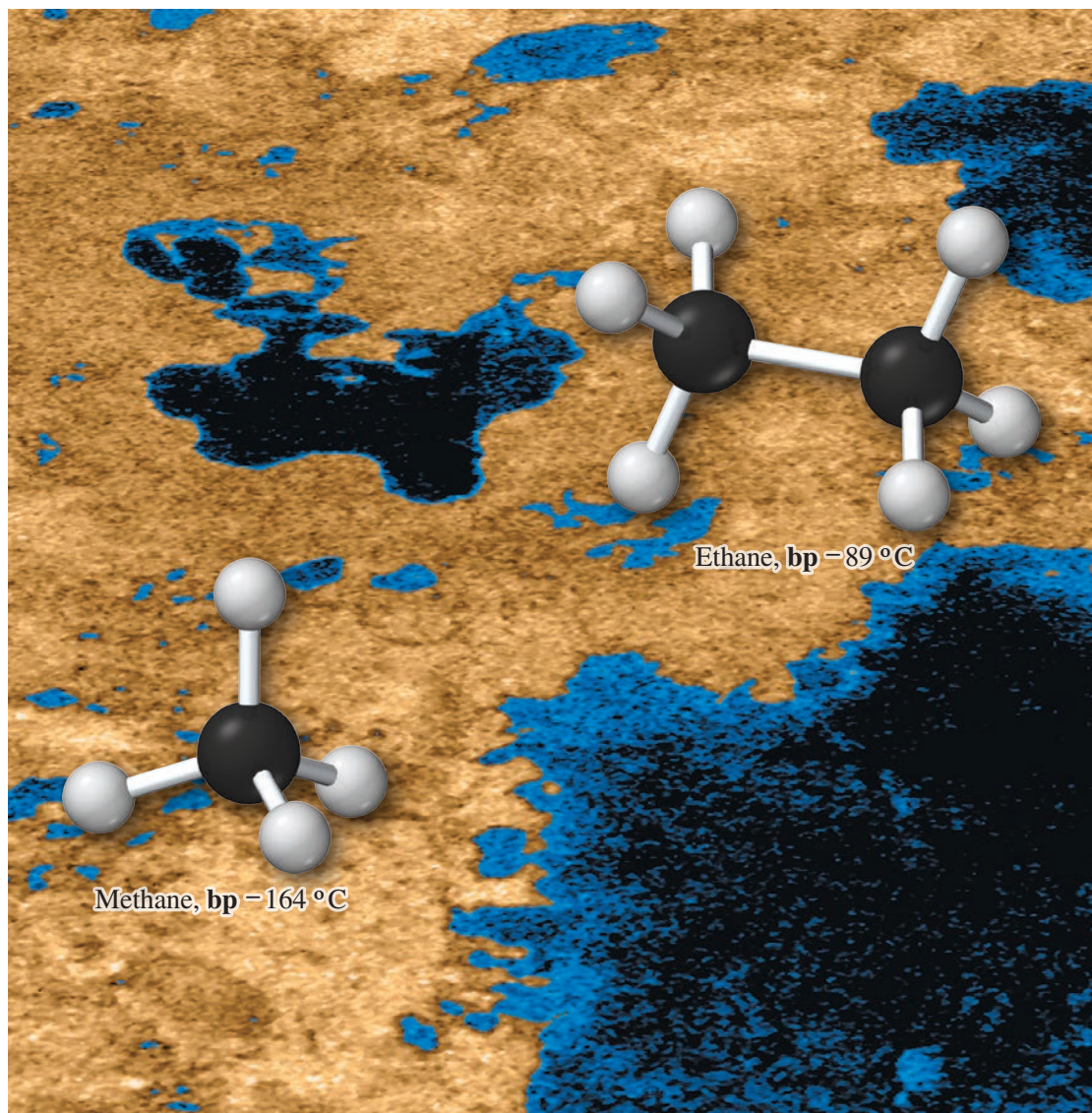
1 Draw and name the isomers of alkanes, and explain the trends in their physical properties.

2 Draw alkane conformations, compare their energies, and predict the most stable conformations.

3 Draw and name the isomers of cycloalkanes, and explain ring strain.

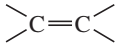
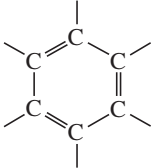
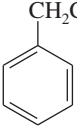
4 Draw the conformations of cycloalkanes, compare their energies, and predict the most stable conformations.

► A colorized radar image of the surface of Titan, Saturn's largest moon, taken by the Cassini spacecraft. The light tan areas correspond to a solid surface with high radar reflectivity. The dark areas of low reflectivity are thought to be lakes of methane and ethane, which rain down on the moon and flow in rivers along its surface.



An **alkane** is a hydrocarbon that contains only single bonds. The alkanes are the simplest and least reactive class of organic compounds because they contain only hydrogen and sp^3 hybridized carbon, and they have no reactive functional groups. Alkanes contain no double or triple bonds and no heteroatoms (atoms other than carbon or hydrogen). They are poor acids and bases, and they are poor electrophiles and nucleophiles as well. Although alkanes undergo reactions such as cracking and combustion at high temperatures, they are much less reactive than other classes of compounds that have functional groups.

TABLE 3-1 Hydrocarbon Classifications

Compound Type	Functional Group	Example
alkanes	none (no double or triple bonds)	$\text{CH}_3\text{—CH}_2\text{—CH}_3$, propane
alkenes	 double bond	$\text{CH}_2=\text{CH—CH}_3$, propene
alkynes	$\text{—C}\equiv\text{C—}$ triple bond	$\text{H—C}\equiv\text{C—CH}_3$, propyne
aromatics	benzene ring 	 ethylbenzene

The study of organic compounds begins with alkanes because alkane structures form the hydrocarbon backbones of most organic compounds. Learning to name organic compounds begins with the names of alkanes because most organic compounds are named using the parent alkane as the root of the name.

3-1 Classification of Hydrocarbons (Review)

We classify hydrocarbons according to their bonding (Section 2-15), as shown in Table 3-1. Alkanes have only single bonds. A hydrocarbon with a carbon–carbon double bond (such as ethylene) is an *alkene*. If a hydrocarbon has a carbon–carbon triple bond (like acetylene), it is an *alkyne*. Hydrocarbons with aromatic rings (resembling benzene) are called *aromatic hydrocarbons*.

A hydrocarbon with no double or triple bonds is said to be **saturated** because it has the maximum number of bonded hydrogens. Another way to describe *alkanes*, then, is as the class of **saturated hydrocarbons**.

3-2 Molecular Formulas of Alkanes

Table 3-2 shows the structures and formulas of the first 20 unbranched alkanes, called the *n*-alkanes or “normal” alkanes. Any isomers of these compounds have the same molecular formulas even though their structures are different. Notice how the molecular formulas increase by two hydrogen atoms each time a carbon atom is added.

The structures of the alkanes in Table 3-2 are purposely written as chains of $\text{—CH}_2\text{—}$ groups (**methylene groups**), terminated at each end by a hydrogen atom. This is the general formula for the unbranched (straight-chain) alkanes. These alkanes differ only by the number of methylene groups in the chain. If the molecule contains n carbon atoms, it must contain $(2n + 2)$ hydrogen atoms. Figure 3-1 shows how this pattern appears in structures and how it leads to formulas of the form $\text{C}_n\text{H}_{2n+2}$.

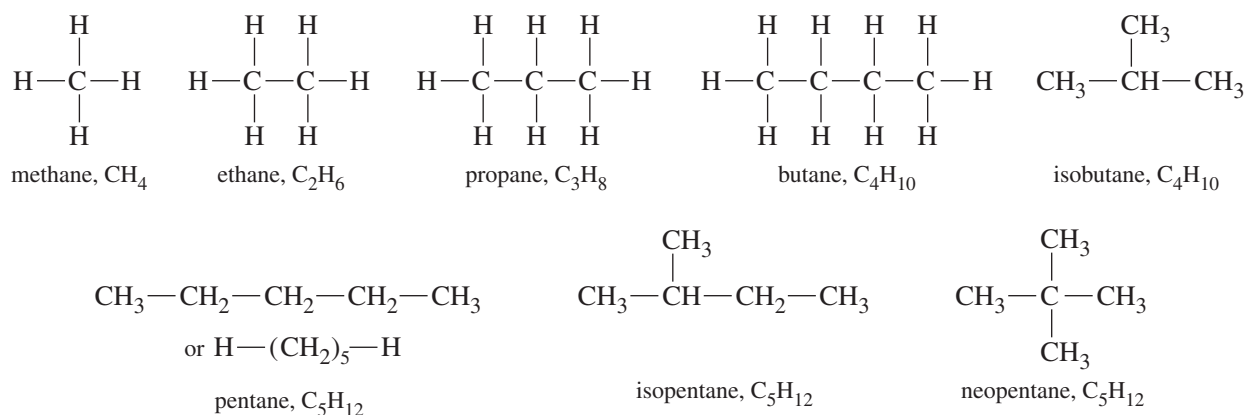
A series of compounds such as the unbranched alkanes, that differ only by the number of $\text{—CH}_2\text{—}$ groups, is called a *homologous series*, and the individual members of the series are called **homologs**. For example, butane is a homolog of propane, and both of these are homologs of hexane and decane.

Although we have derived the $\text{C}_n\text{H}_{2n+2}$ formula using the unbranched *n*-alkanes, it applies to branched alkanes as well. Any isomer of one of these *n*-alkanes has the same molecular formula. Just as butane and pentane follow the $\text{C}_n\text{H}_{2n+2}$ rule, their branched isomers isobutane, isopentane, and neopentane also follow the rule.

TABLE 3-2 Formulas and Physical Properties of the Unbranched Alkanes, Called the *n*-Alkanes

Alkane	Number of Carbons	Structure	Formula	Boiling Point (°C)	Melting Point (°C)	Density ^a
methane	1	H—CH ₂ —H	CH ₄	−164	−183	0.55
ethane	2	H—(CH ₂) ₂ —H	C ₂ H ₆	−89	−183	0.51
propane	3	H—(CH ₂) ₃ —H	C ₃ H ₈	−42	−189	0.50
butane	4	H—(CH ₂) ₄ —H	C ₄ H ₁₀	0	−138	0.58
pentane	5	H—(CH ₂) ₅ —H	C ₅ H ₁₂	36	−130	0.63
hexane	6	H—(CH ₂) ₆ —H	C ₆ H ₁₄	69	−95	0.66
heptane	7	H—(CH ₂) ₇ —H	C ₇ H ₁₆	98	−91	0.68
octane	8	H—(CH ₂) ₈ —H	C ₈ H ₁₈	126	−57	0.70
nonane	9	H—(CH ₂) ₉ —H	C ₉ H ₂₀	151	−51	0.72
decane	10	H—(CH ₂) ₁₀ —H	C ₁₀ H ₂₂	174	−30	0.73
undecane	11	H—(CH ₂) ₁₁ —H	C ₁₁ H ₂₄	196	−26	0.74
dodecane	12	H—(CH ₂) ₁₂ —H	C ₁₂ H ₂₆	216	−10	0.75
tridecane	13	H—(CH ₂) ₁₃ —H	C ₁₃ H ₂₈	235	−5	0.76
tetradecane	14	H—(CH ₂) ₁₄ —H	C ₁₄ H ₃₀	254	6	0.76
pentadecane	15	H—(CH ₂) ₁₅ —H	C ₁₅ H ₃₂	271	10	0.77
hexadecane	16	H—(CH ₂) ₁₆ —H	C ₁₆ H ₃₄	287	18	0.77
heptadecane	17	H—(CH ₂) ₁₇ —H	C ₁₇ H ₃₆	303	23	0.76
octadecane	18	H—(CH ₂) ₁₈ —H	C ₁₈ H ₃₈	317	28	0.76
nonadecane	19	H—(CH ₂) ₁₉ —H	C ₁₉ H ₄₀	330	32	0.78
eicosane	20	H—(CH ₂) ₂₀ —H	C ₂₀ H ₄₂	343	37	0.79
triacontane	30	H—(CH ₂) ₃₀ —H	C ₃₀ H ₆₂	>450	66	0.81

^aDensities are given in g/mL at 20 °C, except for methane and ethane, whose densities are given at their boiling points.

**FIGURE 3-1** Examples of the general alkane molecular formula, C_nH_{2n+2}.

PROBLEM 3-1

Using the general molecular formula for alkanes:

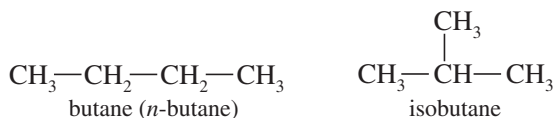
- (a) Predict the molecular formula of the C_{27} straight-chain alkane.
(b) Predict the molecular formula of the alkanes containing 42 carbon atoms with extensive branching.

3-3 Nomenclature of Alkanes

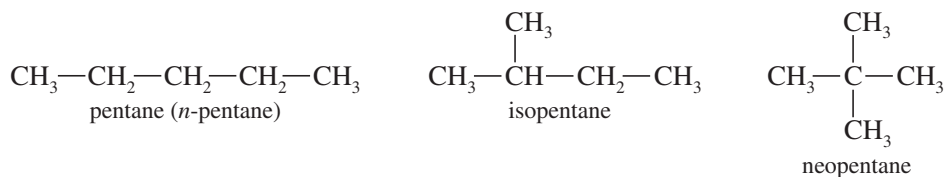
The names *methane*, *ethane*, *propane*, and *butane* have historical roots. From pentane on, alkanes are named using the Greek word for the number of carbon atoms, plus the suffix *-ane* to identify the molecule as an alkane. Table 3-2 gives the names and physical properties of the *n*-alkanes up to 20 carbon atoms.

3-3A Common Names

If all alkanes had unbranched (straight-chain) structures, their nomenclature would be simple. Most alkanes have structural isomers, however, and we need a way of naming all the different isomers. For example, there are two isomers of formula C_4H_{10} . The unbranched isomer is simply called *butane* (or *n-butane*, meaning “normal” butane), and the branched isomer is called *isobutane*, meaning an “isomer of butane.”



The three isomers of C_5H_{12} are called *pentane* (or *n-pentane*), *isopentane*, and *neopentane*.



Isobutane, *isopentane*, and *neopentane* are **common names** or **trivial names**, meaning historical names arising from common usage. Common names cannot easily describe the larger, more complicated molecules having many isomers, however. The number of isomers for any molecular formula grows rapidly as the number of carbon atoms increases. For example, there are 5 structural isomers of hexane, 18 isomers of octane, and 75 isomers of decane! We need a system of nomenclature that enables us to name complicated molecules without having to memorize hundreds of these historical common names.

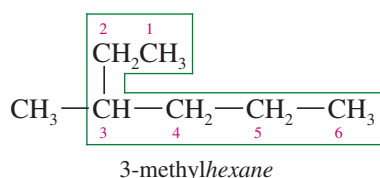
3-3B IUPAC or Systematic Names

A group of chemists representing the countries of the world met in 1892 to devise a system for naming compounds that would be simple to use, require a minimum of memorization, and yet be flexible enough to name even the most complicated organic compounds. This was the first meeting of the group that came to be known as the International Union of Pure and Applied Chemistry, abbreviated **IUPAC**. This international group has developed a detailed system of nomenclature that we call the **IUPAC rules**. The IUPAC rules are accepted throughout the world as the standard method for naming organic compounds. The names that are generated using this system are called **IUPAC names** or **systematic names**. The IUPAC rules for naming organic compounds are summarized (for later reference) in Appendix 5, Summary of Organic Nomenclature.

The IUPAC system works consistently to name many different families of compounds. We will consider the naming of alkanes in detail, and later extend these rules to other kinds of compounds as we encounter them. The IUPAC system uses the longest chain of carbon atoms as the main chain, which is numbered to give the locations of side chains. Four rules govern this process.

Rule 1: The Main Chain The first rule of nomenclature gives the base name of the compound.

Find the longest continuous chain of carbon atoms, and use the name of this chain as the base name of the compound.

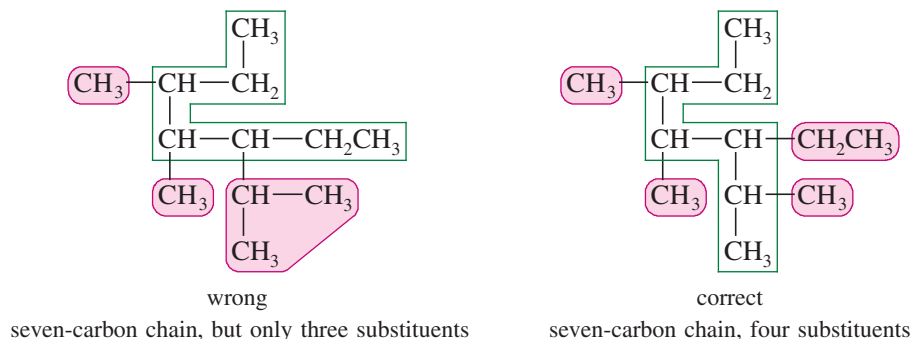


For example, the longest chain of carbon atoms in the compound in the margin contains six carbons, so the compound is named as a *hexane* derivative. The longest chain is rarely drawn in a straight line; look carefully to find it.

The groups attached to the main chain are called **substituents** because they are substituted (in place of a hydrogen atom) on the main chain. *When there are two longest chains of equal length, use the chain with the greater number of substituents as the main chain.* The following compound contains two different seven-carbon chains and is named as a *heptane*. We choose the chain on the right as the main chain because it has more substituents (in red) attached to the chain.

PROBLEM-SOLVING HINT

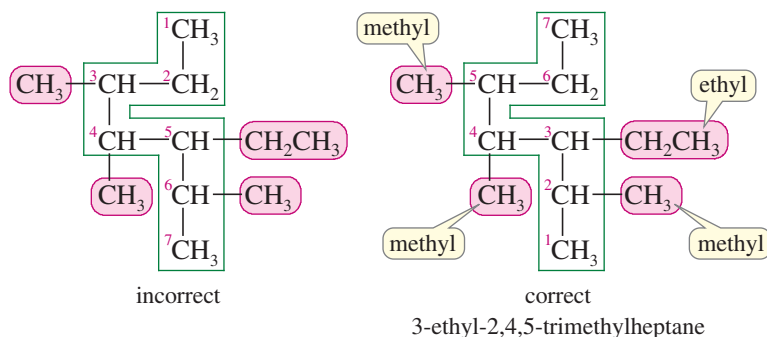
When looking for the longest continuous chain (to give the base name), look for all the different chains of that length. Often, the longest chain with the most substituents is not obvious.



Rule 2: Numbering The Main Chain To give the locations of the substituents, assign a number to each carbon atom on the main chain.

Number the longest chain, beginning with the end of the chain nearest a substituent.

We start the numbering from the end nearest a branch so that the numbers of the substituted carbons will be as low as possible. In the preceding heptane structure on the right, numbering from top to bottom gives the first branch at C3 (carbon atom 3), but numbering from bottom to top gives the first branch at C2. Numbering from bottom to top is correct. (If each end had a substituent the same distance in, we would start at the end nearer the second branch point.)



Rule 3: Naming Alkyl Groups Name the substituent groups.

Name the substituent groups attached to the longest chain as **alkyl groups**. Give the location of each alkyl group by the number of the main-chain carbon atom to which it is attached.

Alkyl groups are named by replacing the *-ane* suffix of the alkane name with *-yl*. *Methane* becomes *methyl*; *ethane* becomes *ethyl*. You may encounter the word *amyl*, which is an archaic term for a pentyl (five-carbon) group.

Alkanes		Alkyl Groups	
CH ₄	methane	CH ₃ —	methyl group
CH ₃ — CH ₃	ethane	CH ₃ — CH ₂ —	ethyl group
CH ₃ — CH ₂ — CH ₃	propane	CH ₃ — CH ₂ — CH ₂ —	propyl group
CH ₃ — (CH ₂) ₂ — CH ₃	butane	CH ₃ — (CH ₂) ₂ — CH ₂ —	butyl group
CH ₃ — (CH ₂) ₃ — CH ₃	pentane	CH ₃ — (CH ₂) ₃ — CH ₂ —	pentyl group (<i>n</i> -amyl group)

The following alkanes show the use of alkyl group nomenclature.

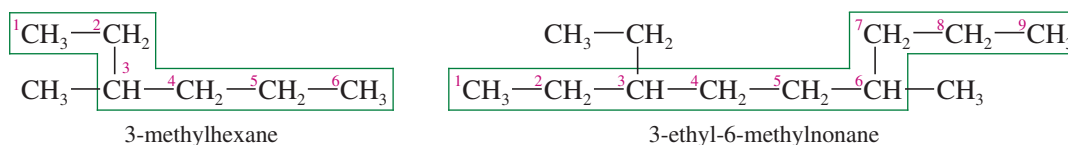
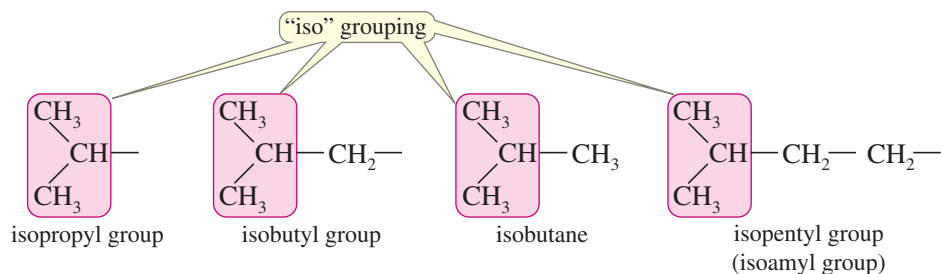


Figure 3-2 gives the names of the most common alkyl groups, those having up to four carbon atoms. The *propyl* and *butyl* groups are simply unbranched three- and four-carbon alkyl groups. These groups are sometimes named “*n*-propyl” and “*n*-butyl” to distinguish them from other kinds of (branched) propyl and butyl groups.

The simple branched alkyl groups are usually known by common names. The isopropyl and isobutyl groups have a characteristic “iso” (CH₃)₂CH grouping, just as in isobutane.



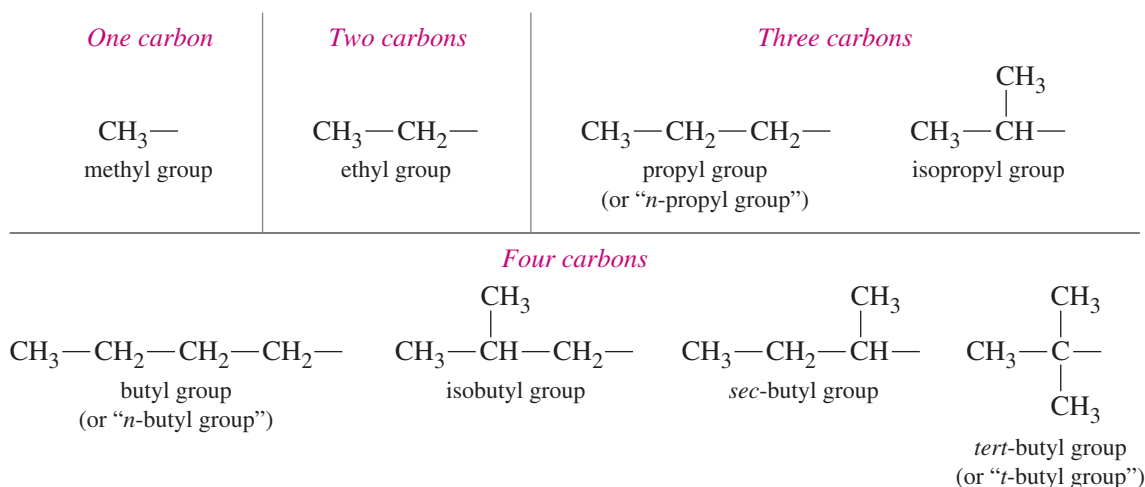
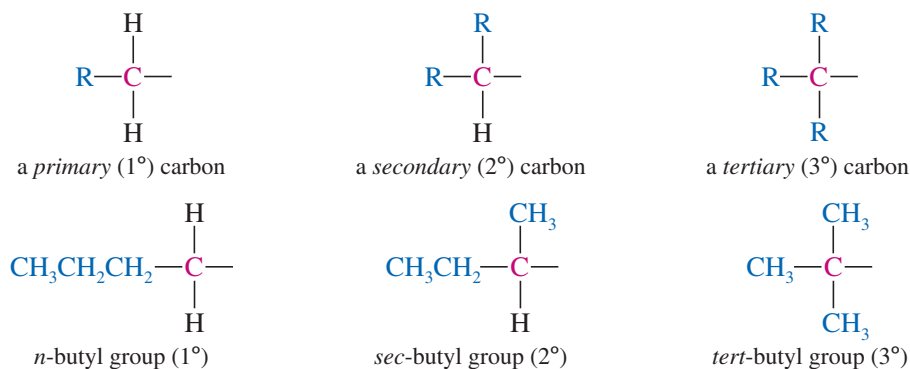


FIGURE 3-2 Some common alkyl groups.

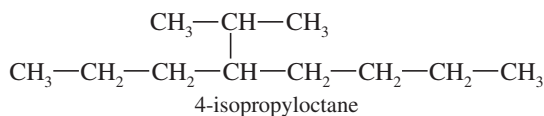
The names of the *secondary*-butyl (*sec*-butyl) and *tertiary*-butyl (*tert*-butyl or *t*-butyl) groups are based on the **degree of alkyl substitution** of the carbon atom attached to the main chain. In the *sec*-butyl group, the carbon atom bonded to the main chain is **secondary** (2°), or bonded to two other carbon atoms. In the *tert*-butyl group, it is **tertiary** (3°), or bonded to three other carbon atoms. In both the *n*-butyl group and the isobutyl group, the carbon atoms bonded to the main chain are **primary** (1°), bonded to only one other carbon atom.

**SOLVED PROBLEM 3-1**

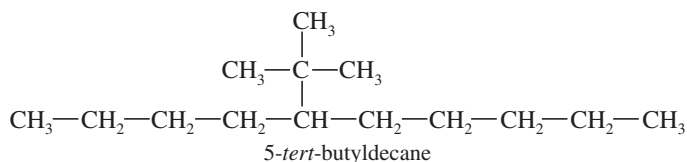
Give the structures of 4-isopropyloctane and 5-*tert*-butyldecane.

SOLUTION

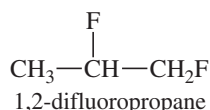
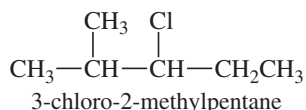
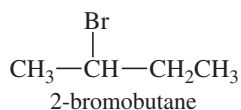
The base name of 4-isopropyloctane is octane, so it has a main chain of eight carbons. There must be an isopropyl group on the fourth carbon from the end.



The base name of 5-*tert*-butyldecane is decane, so it has a main chain of ten carbons. There must be a *tert*-butyl group on the fifth carbon from the end.

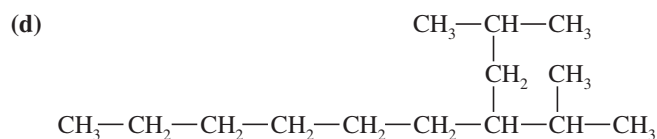
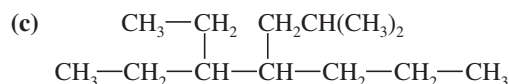
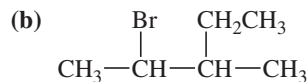
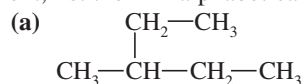


Haloalkanes can be named just like alkanes, with the halogen atom treated as a substituent. Halogen substituents are named *fluoro-*, *chloro-*, *bromo-*, and *iodo-*.



PROBLEM 3-2

Name the following alkanes and haloalkanes. When two or more substituents are present, list them in alphabetical order.



PROBLEM-SOLVING HINT

Please remember the alkyl groups in Figure 3-2. You will encounter them many times throughout this course.

Rule 4: Organizing Multiple Groups The final rule deals with naming compounds with more than one substituent.

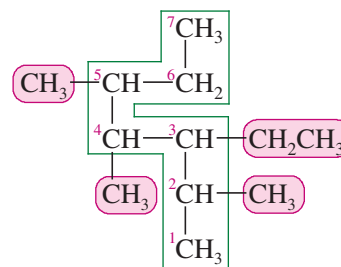
When two or more substituents are present, list them in alphabetical order. When two or more of the *same* alkyl substituent are present, use the prefixes *di-*, *tri-*, *tetra-*, etc., to avoid having to name the alkyl group twice. Include a position number for each substituent, even if it means repeating a number more than once.

di- means 2
tri- means 3
tetra- means 4

penta- means 5
hexa- means 6
hepta- means 7

octa- means 8
nona- means 9
deca- means 10

Using this rule, we can construct names for some complicated structures. Let's finish naming the heptane shown again here in the margin. This compound has an ethyl group on C3 and three methyl groups on C2, C4, and C5. List the ethyl group alphabetically before the methyl groups, and give each of the four substituents a location number.



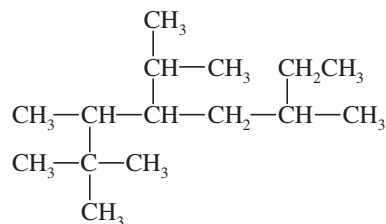
3-ethyl-2,4,5-trimethylheptane

PROBLEM-SOLVING HINT

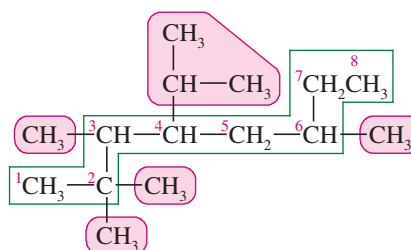
When substituents are alphabetized, *iso-* is used as part of the alkyl group name, but the hyphenated prefixes are not. Thus, *isobutyl* is alphabetized with *i*, but *n*-butyl, *tert*-butyl, and *sec*-butyl are alphabetized with *b*. The number prefixes *di-*, *tri-*, *tetra-*, etc., are ignored in alphabetizing.

SOLVED PROBLEM 3-2

Give a systematic (IUPAC) name for the following compound.

**SOLUTION**

The longest carbon chain contains eight carbon atoms, so this compound is named as an octane. Numbering from left to right gives the first branch on C2; numbering from right to left gives the first branch on C3, so we number from left to right.



There are four methyl groups: two on C2, one on C3, and one on C6. These four groups will be listed as “2,2,3,6-tetramethyl . . .” There is an isopropyl group on C4. Listing the isopropyl group and the methyl groups alphabetically, we have

4-isopropyl-2,2,3,6-tetramethyloctane

SUMMARY Rules for Naming Alkanes

To name an alkane, we follow four rules:

1. Find the longest continuous chain of carbon atoms, and use this chain as the base name.
2. Number the longest chain, beginning with the end nearest a branch.
3. Name the substituents on the longest chain (as alkyl groups). Give the location of each substituent by the number of the main-chain carbon atom to which it is attached.
4. When two or more substituents are present, list them in alphabetical order. When two or more of the *same* alkyl substituent are present, use the prefixes *di-*, *tri-*, *tetra-*, etc. (ignored in alphabetizing) to avoid having to name the alkyl group twice.

PROBLEM-SOLVING HINT

Always compare the total number of carbon atoms in the name with the number in the structure to make sure they match. For example, an *isopropyldimethyloctane* should have $3 + 2 + 8$ carbon atoms.

PROBLEM 3-3

Write structures for the following compounds.

- | | |
|---|---------------------------------------|
| (a) 3-ethyl-4-methylhexane | (b) 3-ethyl-5-isobutyl-3-methylnonane |
| (c) 4- <i>tert</i> -butyl-2-methylheptane | (d) 5-isopropyl-3,3,4-trimethyloctane |

PROBLEM 3-4

Provide IUPAC names for the following compounds.

- (a) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$ (b) $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CH}_3$
- (c) $\begin{array}{c} \text{CH}_2\text{CH}_3 \\ | \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}-\text{CH}(\text{CH}_3)_2 \end{array}$ (d) $\begin{array}{c} \text{CH}_3 \qquad \qquad \text{CH}_2\text{CH}_3 \\ | \qquad \qquad | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$
- (e) $\begin{array}{c} \text{C}(\text{CH}_3)_3 \\ | \\ \text{CH}_3\text{CH}_2\text{CHCHCH}_3 \\ | \\ \text{CH}(\text{CH}_3)_2 \end{array}$ (f) $\begin{array}{c} \text{CH}_3-\text{CHCH}_2\text{CH}_3 \\ | \\ (\text{CH}_3)_3\text{C}-\text{CH}-\text{CH}_2\text{CH}_2\text{CH}_3 \end{array}$

PROBLEM 3-5

All of the following names are incorrect or incomplete. In each case, draw the structure (or a possible structure) and name it correctly.

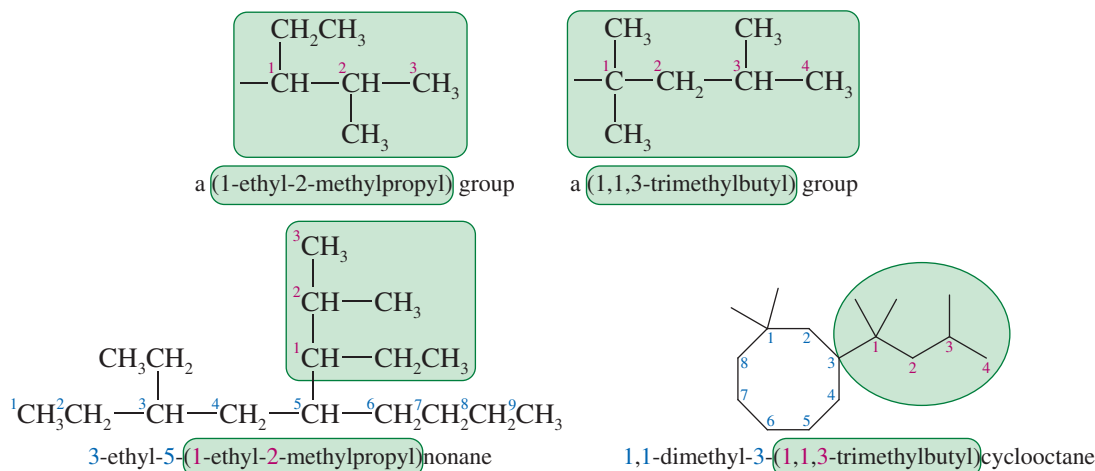
- (a) 3-ethyl-4-methylpentane (b) 2-ethyl-3-methylpentane
- (c) 3-dimethylhexane (d) 4-isobutylheptane
- (e) 2-bromo-3-ethylbutane (f) 2,3-diethyl-5-isopropylheptane

PROBLEM 3-6

Give structures and names for

- (a) the five isomers of C_6H_{14} (b) the nine isomers of C_7H_{16}

Complex Substituents Complex alkyl groups are named by a systematic method using the longest alkyl chain as the base alkyl group. The base alkyl group is numbered beginning with the carbon atom (the “head carbon”) bonded to the main chain. The substituents on the base alkyl group are listed with appropriate numbers, and parentheses are used to set off the name of the complex alkyl group. The following examples illustrate the systematic method for naming complex alkyl groups.

**PROBLEM-SOLVING HINT**

To draw all alkanes of a given molecular formula, start with a straight chain, and then go to shorter chains with more branches. To draw all C_7H_{16} alkanes, for example, start with a C_7 chain, and then go to a C_6 chain with a methyl group in each possible position. Next, draw a C_5 chain with an ethyl group in each possible position, followed by two methyls in all possible combinations. Finally, draw a C_4 chain, which offers only one way to substitute three carbon atoms without extending the longest chain. To guard against duplicated structures, name each compound, and if a name appears twice, you have either duplicated a structure or named something wrong.

PROBLEM 3-7

Draw the structures of the following groups, and give their more common names.

- (a) the (1-methylethyl) group (b) the (2-methylpropyl) group
- (c) the (1-methylpropyl) group (d) the (1,1-dimethylethyl) group
- (e) the (3-methylbutyl) group, sometimes called the “isoamyl” group

PROBLEM 3-8

Draw the structures of the following compounds.

(a) 4-(1,1-dimethylethyl)nonane

(b) 5-(1,2,2-trimethylpropyl)decane

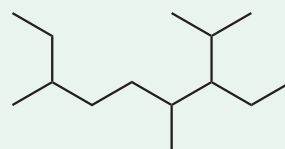
(c) 3,3-diethyl-4-(2,2-dimethylpropyl)nonane

PROBLEM 3-9

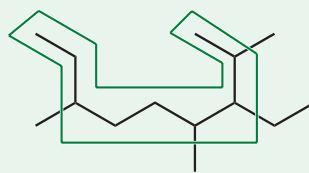
Without looking at the structures, give molecular formulas for the compounds in Problem 3-8 (a) and (b). Use the names of the groups to determine the number of carbon atoms; then use the $(2n+2)$ rule.

FOCUS Applying Nomenclature Rules

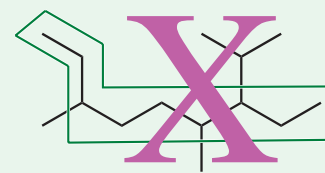
Apply the rules of IUPAC nomenclature to name this compound:



Rule 1: Find the longest continuous carbon chain; name it. If there are two chains of equal length, use the chain with the greater number of substituents.

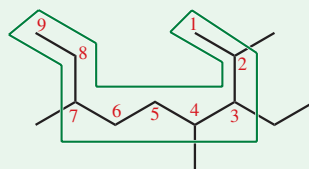


9 carbon chain = nonane
4 substituents

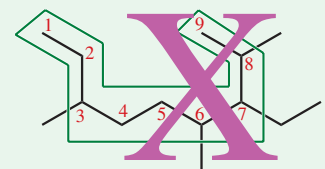


9 carbon chain = nonane
only 3 substituents

Rule 2: Number the longest chain, beginning with the end nearest a branch.

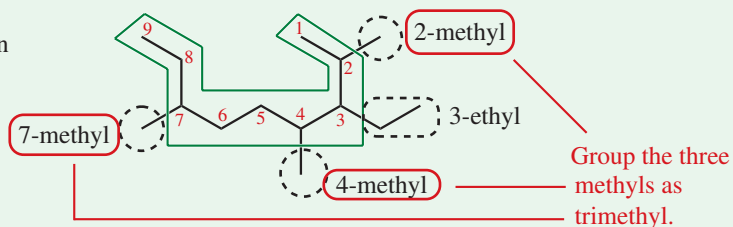


first substituent on C2



first substituent on C3

Rule 3: Write the name and location number of each substituent.



Rule 4:

- Alphabetize substituents: *ethyl* before *methyl*. In alphabetizing, ignore *di*, *tri*, etc., but include *iso* and *cyclo*.
- Group any of the same substituents: 2,4,7-trimethyl
- Place substituent names before the base name. Separate numbers from letters with hyphens; separate numbers with commas.

The correct name is: **3-ethyl-2,4,7-trimethylnonane**

3-4 Physical Properties of Alkanes

Alkanes are used primarily as fuels, solvents, and lubricants. Natural gas, gasoline, kerosene, heating oil, lubricating oil, and paraffin “wax” are all composed primarily of alkanes, with different physical properties resulting from different ranges of molecular weights.

3-4A Solubilities and Densities of Alkanes

Alkanes are nonpolar, so they dissolve in nonpolar or weakly polar organic solvents. Alkanes are said to be **hydrophobic** (“water-hating”) because they do not dissolve in water. Alkanes are good lubricants and preservatives for metals because they keep water from reaching the metal surface and causing corrosion.

Densities of the *n*-alkanes are listed in Table 3-2 (page 148). Alkanes have densities around 0.7 g/mL, compared with a density of 1.0 g/mL for water. Because alkanes are less dense than water and insoluble in water, a mixture of an alkane (such as gasoline or oil) and water quickly separates into two phases, with the alkane on top.

3-4B Boiling Points of Alkanes

Table 3-2 also gives the boiling points and melting points of the unbranched alkanes. The boiling points increase smoothly with increasing numbers of carbon atoms and increasing molecular weights. Larger molecules have larger surface areas, resulting in increased intermolecular van der Waals attractions. These increased attractions must be overcome for vaporization and boiling to occur. Thus, a larger molecule, with greater surface area and greater van der Waals attractions, boils at a higher temperature.

A graph of *n*-alkane boiling points versus the number of carbon atoms (the blue line in Figure 3-3) shows that boiling points increase with increasing molecular weight. Each additional CH₂ group increases the boiling point by about 30 °C up to about ten carbons, and by about 20 °C in higher alkanes.

The green line in Figure 3-3 represents the boiling points of some branched alkanes. In general, a branched alkane boils at a lower temperature than the *n*-alkane with the same number of carbon atoms. This difference in boiling points arises because branched alkanes are more compact, with less surface area for London force interactions.

3-4C Melting Points of Alkanes

The blue line in Figure 3-4 is a graph of the melting points of the *n*-alkanes. Like their boiling points, the melting points increase with increasing molecular weight. The melting point graph is not smooth, however. Alkanes with even numbers of carbon atoms pack better into a solid structure, so higher temperatures are needed to melt them. Alkanes with odd numbers of carbon atoms do not pack as well, and they melt at lower temperatures. The sawtooth-shaped graph of melting points is smoothed by drawing separate lines (green and red) for the alkanes with even and odd numbers of carbon atoms.



Oil floats on water. Note how the oil (from the 2010 blowout of the Macondo well in the Gulf of Mexico) floats on top of the water, where it can be burned off. Oil recovery booms, containing nonpolar fibers, are used to soak up and contain the floating oil. Note how most of the burning oil ends at the oil recovery boom.

FIGURE 3-3 Alkane boiling points. The boiling points of the unbranched alkanes (blue) are compared with those of some branched alkanes (green). Because of their smaller surface areas, branched alkanes have lower boiling points than unbranched alkanes.

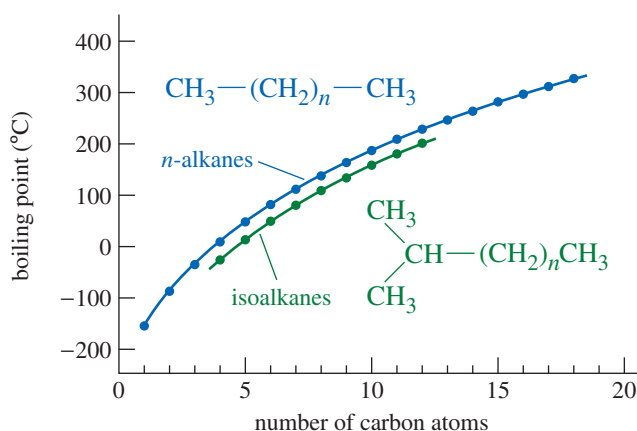
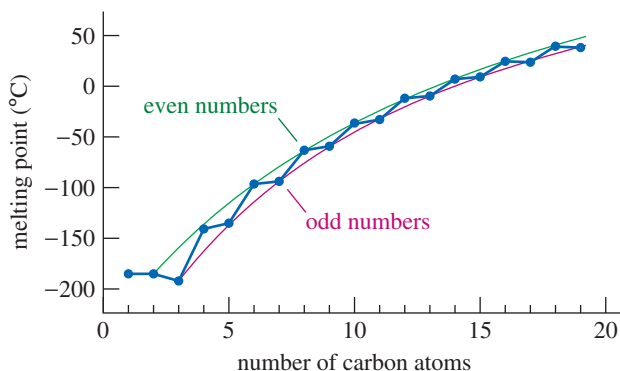


FIGURE 3-4 Alkane melting points. The melting-point curve for n -alkanes with even numbers of carbon atoms is slightly higher than the curve for alkanes with odd numbers of carbons.



PROBLEM 3-10

List each set of compounds in order of increasing boiling point.

- (a) hexane, octane, and decane
 (b) octane, $(\text{CH}_3)_3\text{C}-\text{C}(\text{CH}_3)_3$, and $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}_2\text{CH}_3$

3-5 Uses and Sources of Alkanes

Distillation of petroleum separates alkanes into fractions with similar boiling points. These fractions are suited for different uses based on their physical properties, such as volatility and viscosity.

3-5A Major Uses of Alkanes

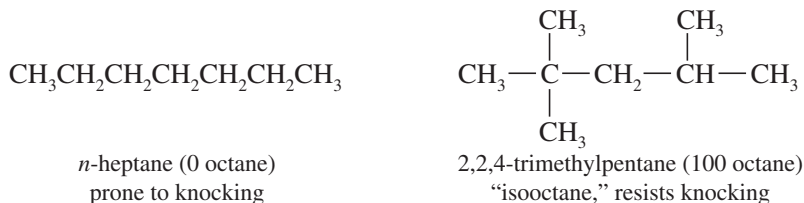
C₁–C₂ Methane and ethane are gases at room temperature and atmospheric pressure. Methane is the major constituent of *natural gas*, with a variable amount of ethane present as well. Natural gas is widely used as a clean-burning fuel for heating and power generation.

Both methane and ethane are difficult to liquefy, so they are usually handled as compressed gases. Upon cooling to cryogenic (very low) temperatures, however, methane and ethane become liquids. *Liquefied natural gas*, mostly methane, can be transported in special refrigerated tankers more easily than it can be transported as a compressed gas.

C₃–C₄ Propane and butane are also gases at room temperature and pressure, but they are easily liquefied at room temperature under modest pressure. These gases, often obtained along with liquid petroleum, are stored in low-pressure cylinders of *liquefied petroleum gas (LPG)*. Propane and butane are good fuels, both for heating and for internal combustion engines. They burn cleanly, and pollution-control equipment is rarely necessary. In many agricultural areas, propane and butane are more cost-effective tractor fuels than gasoline and diesel fuel. Propane and butane have largely replaced freons (see Section 6-3D) as propellants in aerosol cans. Unlike alkanes, the chlorofluorocarbon Freon propellants are implicated in damaging Earth's protective ozone layer.

C₅–C₈ The next four alkanes are free-flowing, volatile liquids. Isomers of pentane, hexane, heptane, and octane are the primary constituents of gasoline. Their volatility is crucial for this use because the injection system simply squirts a stream of gasoline into the intake air as it rushes through. If gasoline did not evaporate easily, it would reach the cylinder in the form of droplets. Droplets cannot burn as efficiently as a vapor, so the engine would smoke and give low mileage.

In addition to being volatile, gasoline must resist the potentially damaging explosive combustion known as *knocking*. The antiknock properties of gasoline are rated by an **octane number** that is assigned by comparing the gasoline to a mixture of *n*-heptane (which knocks badly) and isooctane (2,2,4-trimethylpentane, which is not prone to knocking). The gasoline being tested is used in a test engine with a variable compression ratio. Higher compression ratios induce knocking, so the compression ratio is increased until knocking begins. Tables are available that show the percentage of isooctane in an isooctane/heptane blend that begins to knock at any given compression ratio. The octane number assigned to the gasoline is simply the percentage of isooctane in an isooctane/heptane mixture that begins to knock at that same compression ratio.



C₉–C₁₆ The nonanes (C₉) through about the hexadecanes (C₁₆) are higher-boiling liquids that are somewhat viscous. These alkanes are used in kerosene, jet fuel, and diesel fuel. **Kerosene**, the lowest-boiling of these fuels, is less volatile than gasoline and less prone to forming explosive mixtures. Kerosene has been used in kerosene lamps and heaters, which use wicks to allow this heavier fuel to burn. Jet fuel is similar to kerosene, but more highly refined and less odorous.

Diesel fuel is not very volatile, so it does not evaporate in the intake air entering an engine. In a diesel engine, the fuel is sprayed directly into the cylinder right at the top of the compression stroke. The hot, highly compressed air in the cylinder causes the fuel to burn quickly, swirling and vaporizing as it burns. Some of the alkanes in diesel fuel have fairly high freezing points, and they may solidify in cold weather. This partial solidification causes the diesel fuel to turn into a waxy, semisolid mass. Owners of diesel engines in cold climates often mix a small amount of gasoline with their diesel fuel in the winter. The added gasoline dissolves the frozen alkanes, diluting the slush and allowing it to be pumped to the cylinders.

C₁₆ and Up Alkanes with more than 16 carbon atoms are most often used as lubricating and heating oils. These are sometimes called "mineral" oils because they come from petroleum, which was once considered a mineral.



Clean-burning vehicles powered by natural gas help to reduce air pollution in urban areas.

Application: Fuels

Tetraethyl lead (TEL), formula (CH₃CH₂)₄Pb, was once added to gasoline to increase the octane rating and lubricate the valves. In the 1970s, lead was banned from automotive gasoline because it inactivates catalytic converters and introduces lead into the environment. TEL is still used in 100LL, which is low-lead 100-octane aviation fuel for piston aircraft engines. No suitable replacement fuel has yet been certified for the old engines.

Application: Air Pollution

Incomplete combustion of gasoline and other motor fuels releases significant quantities of volatile organic compounds (VOCs) into the atmosphere. VOCs are composed of short-chained alkanes, alkenes, aromatic compounds, and a variety of other hydrocarbons. VOCs are components of air pollution and contribute to cardiac and respiratory diseases.



The large distillation tower at left is used to separate petroleum into fractions based on their boiling points. The “cat cracker” at right uses catalysts and high temperatures to crack large molecules into smaller ones.

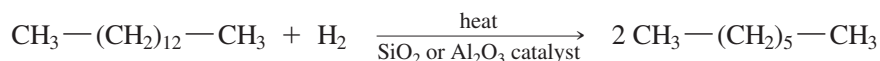
Paraffin “wax” is not a true wax, but a purified mixture of high-molecular-weight alkanes with melting points well above room temperature. The true waxes are long-chain esters, discussed in Chapter 25.

3-5B Alkane Sources; Petroleum Refining

Alkanes are derived mostly from petroleum and petroleum by-products. *Petroleum*, often called *crude oil*, is pumped from wells that reach into pockets containing the remains of prehistoric plants. The principal constituents of crude oil are alkanes, some aromatics, and some undesirable compounds containing sulfur and nitrogen. The composition of petroleum and the amounts of contaminants vary from one source to another, and a refinery must be carefully adjusted to process a particular type of crude oil. Because of their different qualities, different prices are paid for light Arabian crude, West Texas crude, and other classes of crude petroleum.

The first step in refining petroleum is a careful fractional distillation. The products of that distillation are not pure alkanes, but mixtures of alkanes with useful ranges of boiling points. Table 3-3 shows the major fractions obtained from the distillation of crude petroleum.

After distillation, **catalytic cracking** converts some of the less valuable fractions to more valuable products. Catalytic cracking involves heating alkanes in the presence of materials that catalyze the cleavage of large molecules into smaller ones. Cracking is often used to convert higher-boiling fractions into mixtures that can be blended with gasoline. When cracking is done in the presence of hydrogen (**hydrocracking**), the product is a mixture of alkanes free of sulfur and nitrogen impurities. The following reaction shows the catalytic hydrocracking of a molecule of tetradecane into two molecules of heptane.



3-5C Natural Gas; Methane

Natural gas was once treated as a waste product of petroleum production and destroyed by flaring it off. Now natural gas is an equally valuable natural resource, pumped and stored throughout the world. Natural gas is about 70% methane, 10% ethane, and 15% propane, depending on the source of the gas. Small amounts of other hydrocarbons and contaminants are also present. Natural gas is often found above pockets of petroleum or coal, although it is also found in places where there is little or no recoverable petroleum or coal. Natural gas is used primarily as a fuel to heat buildings and to generate electricity. It is also important as a starting material for the production of fertilizers.

TABLE 3-3 Major Fractions Obtained from Distillation of Crude Petroleum

Boiling Range (°C)	Number of Carbons	Fraction	Use
under 30°	2–4	petroleum gas	LP gas for heating
30°–180°	4–9	gasoline	motor fuel
160°–230°	8–16	kerosene	heating, jet fuel
200°–320°	10–18	diesel	motor fuel
300°–450°	16–30	heavy oil	heating, lubrication
>300° (vacuum)	>25	petroleum “jelly,” paraffin “wax”	
residue	>35	asphalt	

Although the methane we burn as natural gas is millions of years old, another 300 million tons per year (estimated) of new methane is synthesized by microbes in diverse places such as the stomachs of plant-eating animals and the mud under the seafloor. Most of the undersea methane is eaten by other microbes, but some escapes at methane seeps. Under the sea, cold, high-pressure conditions may allow formation of **methane hydrate**, with individual methane molecules trapped inside cages of water molecules. When methane hydrate is brought to the surface, it quickly melts and the methane escapes. Methane is sometimes generated on a small scale from manure and landfill waste, but we have no practical methods for capturing and using naturally-occurring microbial methane or methane hydrate. Much of this methane escapes to the atmosphere, where it acts as a greenhouse gas and contributes to global warming.

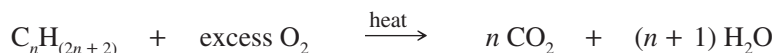
3-6 Reactions of Alkanes

Alkanes are the least reactive class of organic compounds. Their low reactivity is reflected in another term for alkanes: **paraffins**. The name *paraffin* comes from two Latin terms, *parum*, meaning “too little,” and *affinis*, meaning “affinity.” Chemists found that alkanes do not react with strong acids or bases or with most other reagents. They attributed this low reactivity to a lack of affinity for other reagents, so they coined the name “paraffins.”

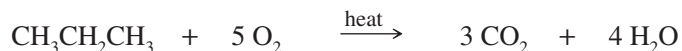
Most useful reactions of alkanes take place under energetic or high-temperature conditions. These conditions are inconvenient in a laboratory because they require specialized equipment, and the rate of the reaction is difficult to control. Alkane reactions often form mixtures of products that are difficult to separate. These mixtures may be commercially important for an industry, however, where the products may be separated and sold separately. Newer methods of selective functionalization may eventually change this picture. For now, however, the following alkane reactions are rarely seen in laboratory applications, but they are widely used in the chemical industry and even in your home and car.

3-6A Combustion

Combustion is a rapid oxidation that takes place at high temperatures, converting alkanes to carbon dioxide and water. Little control over the reaction is possible, except for moderating the temperature and controlling the fuel/air ratio to achieve efficient burning.



Example



Unfortunately, the burning of gasoline and fuel oil pollutes the air and depletes the petroleum resources needed for lubricants and chemical feedstocks. Solar and nuclear heat sources cause less pollution, and they do not deplete these important natural resources. Facilities that use these more environment-friendly heat sources are currently more expensive than those that rely on the combustion of alkanes.

3-6B Cracking and Hydrocracking

As discussed in Section 3-5B, catalytic **cracking** of large hydrocarbons at high temperatures produces smaller hydrocarbons. The cracking process usually operates under conditions that give the maximum yields of gasoline. In **hydrocracking**, hydrogen is added to give saturated hydrocarbons; cracking without hydrogen gives mixtures of alkanes and alkenes.



Methane hydrate, consisting of methane molecules surrounded by water molecules, is formed under high pressure on the cold seafloor. When brought to the surface, it quickly melts and releases the methane. In this photo, the methane has been ignited to make it visible.

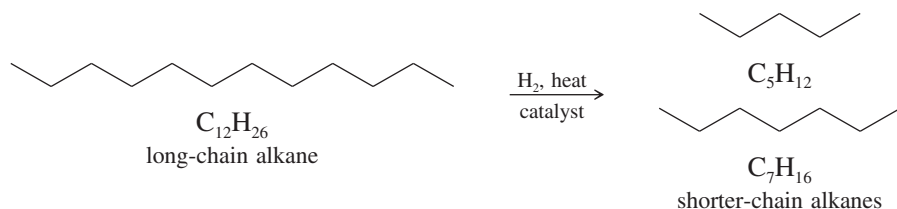
Application: Global Warming

Methane is a stronger greenhouse gas than carbon dioxide. If all the methane trapped in methane hydrates were suddenly released into the atmosphere (by warming of the ocean, for example), the rate of global warming could increase dramatically.

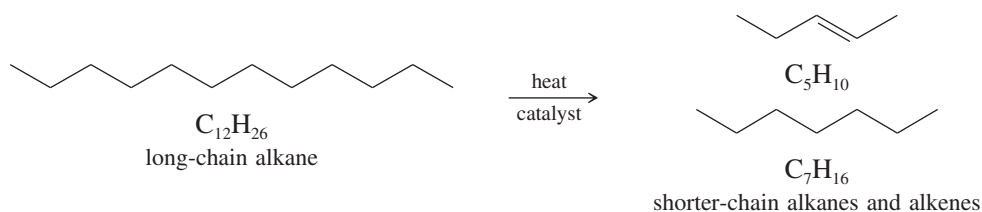


Combustion is the most common reaction of alkanes. Lightning initiated this fire in a tank containing 3 million gallons of gasoline at the Shell Oil storage facility in Woodbridge, New Jersey (June 11, 1996).

Catalytic hydrocracking

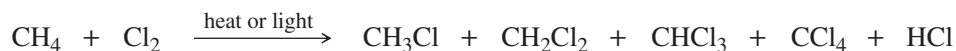


Catalytic cracking



3-6C Halogenation

Alkanes can react with halogens (F_2 , Cl_2 , Br_2 , I_2) to form alkyl halides. For example, methane reacts with chlorine (Cl_2) to form chloromethane (methyl chloride), dichloromethane (methylene chloride), trichloromethane (chloroform), and tetrachloromethane (carbon tetrachloride).



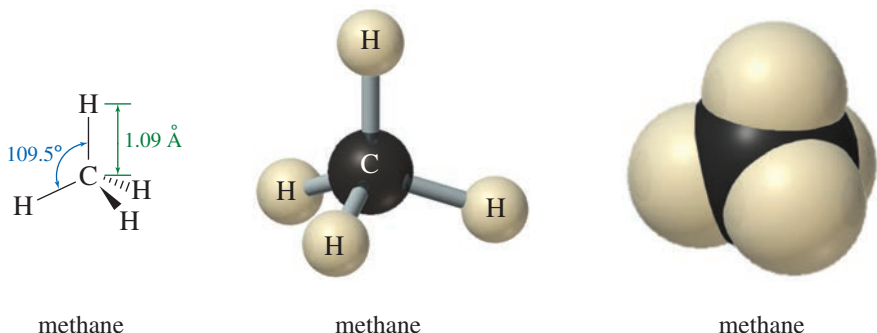
Heat or light is usually needed to initiate this **halogenation**. Reactions of alkanes with chlorine and bromine proceed at moderate rates and are easily controlled. Reactions with fluorine are often too fast to control, however. Iodine reacts very slowly or not at all. We will discuss the halogenation of alkanes in Chapter 4.

3-7 Structure and Conformations of Alkanes

Although alkanes are not as reactive as other classes of organic compounds, they have many of the same structural characteristics. We will use simple alkanes as examples to study some of the properties of organic compounds, including the structure of sp^3 hybridized carbon atoms and properties of $\text{C}-\text{C}$ and $\text{C}-\text{H}$ single bonds.

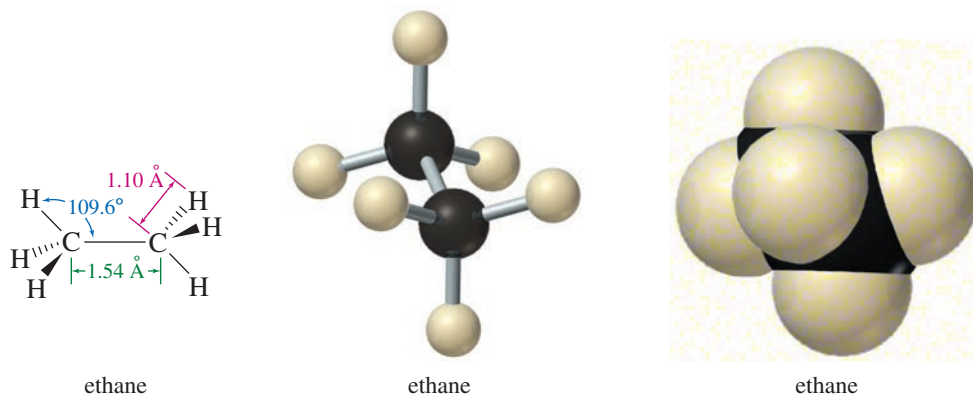
3-7A Structure of Methane

The simplest alkane is *methane*, CH_4 . Methane is perfectly tetrahedral, with the 109.5° bond angles predicted for an sp^3 hybridized carbon. Four hydrogen atoms are covalently bonded to the central carbon atom, with bond lengths of $1.09 \text{ \AA}$.

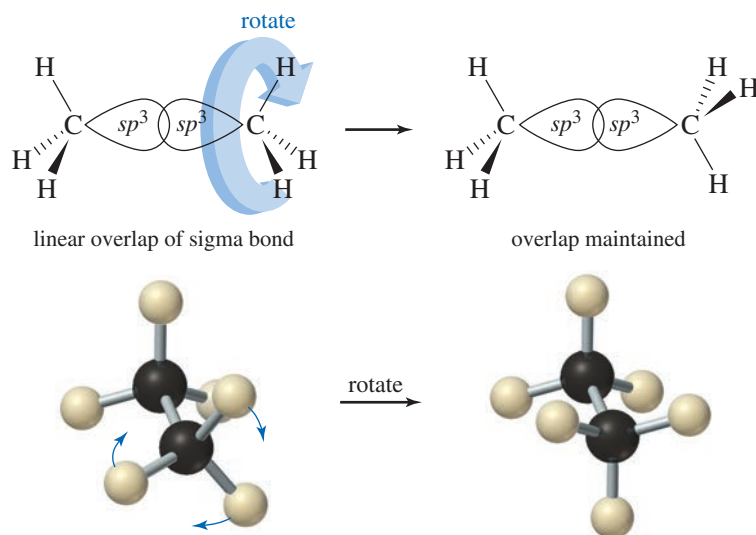


3-7B Conformations of Ethane

Ethane, the two-carbon alkane, is composed of two methyl groups with overlapping sp^3 hybrid orbitals forming a sigma bond between them.



The two methyl groups are not fixed in a single position, but are relatively free to rotate about the sigma bond connecting the two carbon atoms. The bond maintains its linear bonding overlap as the carbon atoms turn. The different arrangements formed by rotations about a single bond are called **conformations**, and a specific conformation is called a **conformer** ("conformational isomer").* Pure conformers cannot be isolated in most cases, because the molecules are constantly rotating through all the possible conformations.



In drawing conformations, we often use **Newman projections**, a way of drawing a molecule looking straight down the bond connecting two carbon atoms (Figure 3-5). The front carbon atom is represented by three lines (three bonds) coming together in a Y shape. The back carbon is represented by a circle with three bonds pointing out from it. Until you become familiar with Newman projections, you should make models and compare your models with the drawings.

An infinite number of conformations are possible for ethane, because the angle between the hydrogen atoms on the front and back carbons can take on an infinite number of values. Figure 3-6 uses Newman projections and sawhorse structures to

*This is the common definition of conformers. The IUPAC definition also requires that a conformer correspond to a distinct potential energy minimum, such as the anti and gauche conformations of butane.

FIGURE 3-5 The Newman projection looks straight down the carbon–carbon bond.

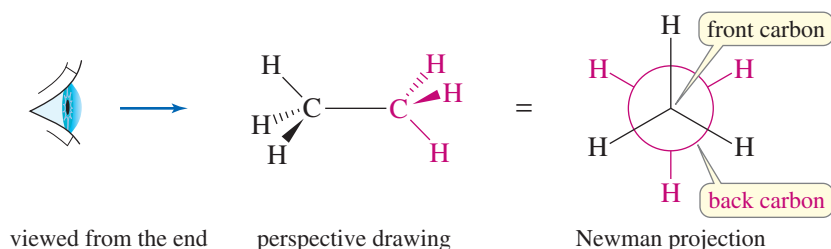
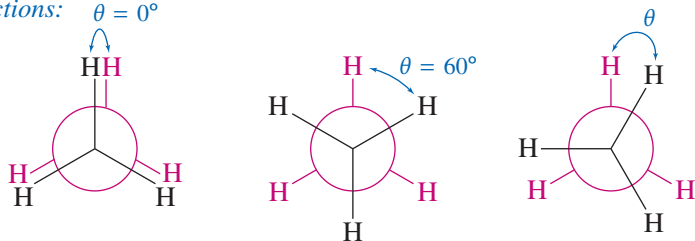
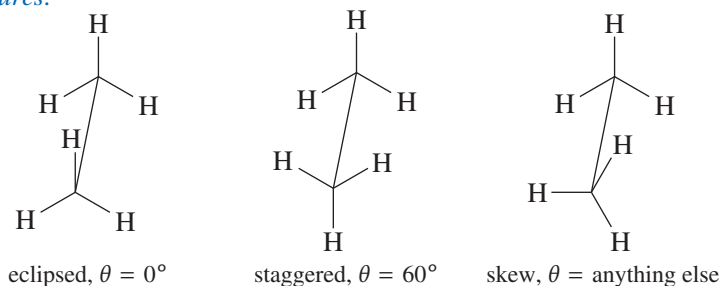


FIGURE 3-6 Ethane conformations. The eclipsed conformation has a dihedral angle $\theta = 0^\circ$, and the staggered conformation has $\theta = 60^\circ$. Any other conformation is called a skew conformation.

Newman projections: $\theta = 0^\circ$

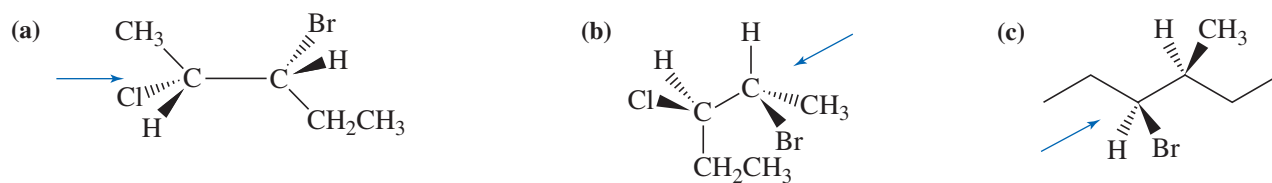


Sawhorse structures:



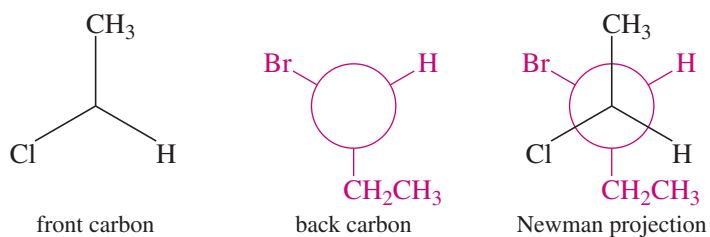
PROBLEM 3-11 (PARTIALLY SOLVED)

Draw Newman projections of the following molecules viewed from the direction of the blue arrows.



SOLUTION (a)

Looking from the direction of the arrow, the front carbon has a methyl group pointing straight up, a chlorine pointing down and to the left, and a hydrogen pointing down and to the right. The back carbon has an ethyl group pointing straight down, a bromine pointing up and to the left, and a hydrogen pointing up and to the right.



illustrate some of these ethane conformations. **Sawhorse structures** picture the molecule looking down at an angle toward the carbon–carbon bond. Sawhorse structures can be misleading, depending on how the eye sees them. We will generally use perspective or Newman projections to draw molecular conformations.

Any conformation can be specified by its **dihedral angle** (θ), the angle between the C—H bonds on the front carbon atom and the C—H bonds on the back carbon in the Newman projection. Two of the conformations have special names. The conformation with $\theta = 0^\circ$ is called the **eclipsed conformation** because the Newman projection shows the hydrogen atoms on the back carbon to be hidden (eclipsed) by those on the front carbon. The **staggered conformation**, with $\theta = 60^\circ$, has the hydrogen atoms on the back carbon staggered halfway between the hydrogens on the front carbon. Any other intermediate conformation is called a **skew conformation**.

In a sample of ethane gas at room temperature, the ethane molecules rotate millions of times per second, and their conformations are constantly changing. These conformations are not all equally favored, however. The lowest-energy conformation is the staggered conformation, with the electron clouds in the C—H bonds separated as much as possible. The interactions of the electrons in the bonds make the eclipsed conformation about 12.6 kJ/mol (3.0 kcal/mol) higher in energy than the staggered conformation. Three kilocalories is not a large amount of energy, and at room temperature, most molecules have enough kinetic energy to overcome this small rotational barrier.

Figure 3-7 shows how the potential energy of ethane changes as the carbon–carbon bond rotates. The y axis shows the potential energy relative to the most stable (staggered) conformation. The x axis shows the dihedral angle as it increases from 0° (eclipsed) through 60° (staggered) and on through additional eclipsed and staggered conformations as θ continues to increase. As ethane rotates toward an eclipsed conformation, its potential energy increases, and there is resistance to the rotation. This resistance to twisting (torsion) is called **torsional strain**, and the 12.6 kJ/mol (3.0 kcal/mol) of energy required is called **torsional energy**.

Conformational analysis is the study of the energetics of different conformations. Many reactions depend on a molecule's ability to twist into a particular conformation; conformational analysis can help to predict which conformations are favored and which reactions are more likely to take place. We will apply conformational analysis to propane and butane first, and later to some interesting cycloalkanes.

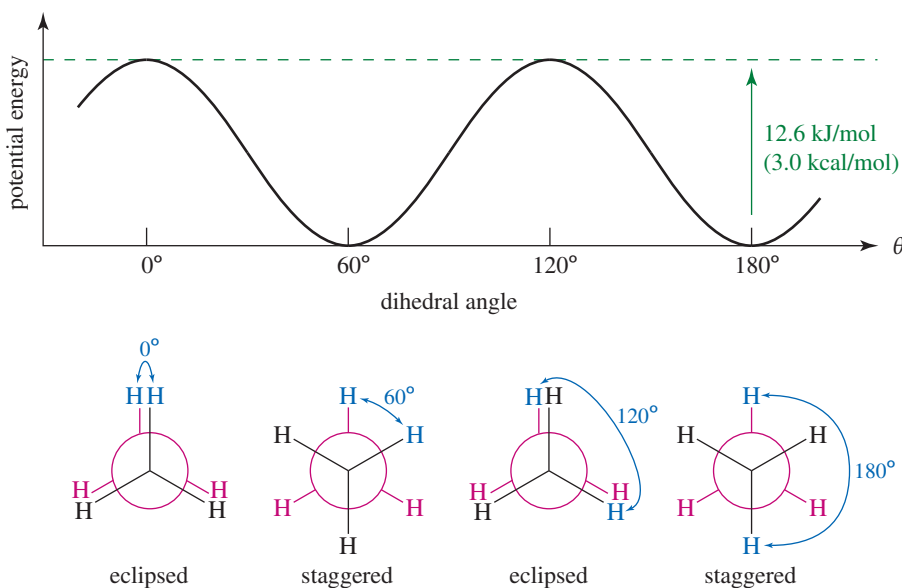


FIGURE 3-7 The torsional energy of ethane is lowest in the staggered conformation. The eclipsed conformation is about 12.6 kJ/mol (3.0 kcal/mol) higher in energy. At room temperature, this barrier is easily overcome and the molecules rotate constantly.

FIGURE 3-8 Propane is shown here as a perspective drawing and as a Newman projection looking down one of the carbon–carbon bonds.

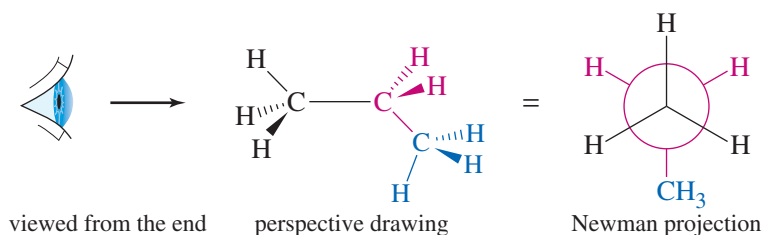
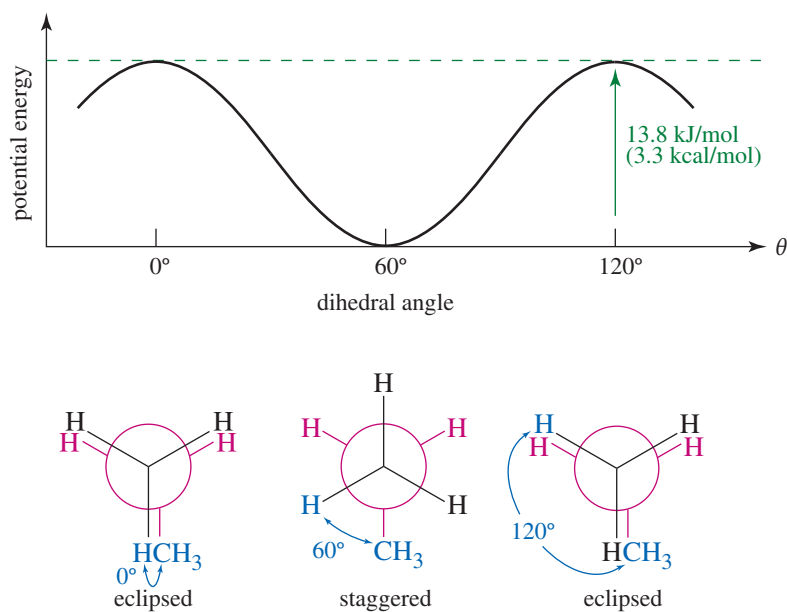


FIGURE 3-9 Torsional energy of propane. When a C—C bond of propane rotates, the torsional energy varies much like it does in ethane, but with 13.8 kJ/mol (3.3 kcal/mol) torsional energy in the eclipsed conformation.



3-7C Conformations of Propane

Propane is the three-carbon alkane, with formula C_3H_8 . Figure 3-8 shows a three-dimensional representation of propane and a Newman projection looking down one of the carbon–carbon bonds.

Figure 3-9 shows a graph of the torsional energy of propane as one of the carbon–carbon bonds rotates. The torsional energy of the eclipsed conformation is about 13.8 kJ/mol (3.3 kcal/mol), only 1.2 kJ (0.3 kcal) more than that required for ethane. Apparently, the torsional strain resulting from eclipsing a carbon–hydrogen bond with a carbon–methyl bond is only 1.2 kJ (0.3 kcal) more than the strain of eclipsing two carbon–hydrogen bonds.

PROBLEM-SOLVING HINT

A C—H bond eclipsed with another C—H bond contributes 4.2 kJ/mol (1.0 kcal/mol) torsional energy (one-third of eclipsed ethane). A C—H bond eclipsed with a C—CH₃ bond contributes 5.4 kJ/mol (1.3 kcal/mol).

PROBLEM 3-12

Draw a graph, similar to Figure 3-9, of the torsional strain of 2-methylpropane as it rotates about the bond between C1 and C2. Show the dihedral angle and draw a Newman projection for each staggered and eclipsed conformation.

3-8 Conformations of Butane

Butane is the four-carbon alkane, with molecular formula C_4H_{10} . We refer to *n*-butane as a straight-chain alkane, but the chain of carbon atoms is not really straight. The angles between the carbon atoms are close to the tetrahedral angle, about 109.5°.

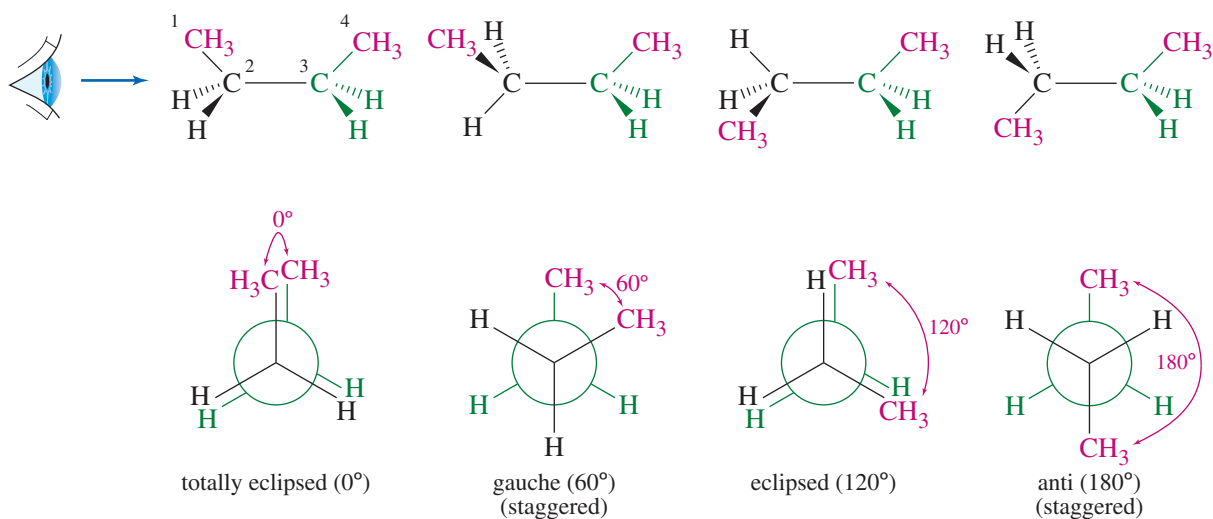


FIGURE 3-10 Butane conformations. Rotations about the center bond in butane give different molecular shapes. Three of these conformations have specific names.

Rotations about any of the carbon–carbon bonds are possible. Rotations about either of the end bonds (C1—C2 or C3—C4) just rotate a methyl group like they do in ethane and propane. Rotations about the central C2—C3 bond are more interesting, however. Figure 3-10 shows Newman projections, looking along the central C2—C3 bond, for four conformations of butane. Construct butane with your molecular models, and sight down the C2—C3 bond. Notice that we have defined the dihedral angle θ as the angle between the two end methyl groups.

Three of the conformations shown in Figure 3-10 are given special names. When the methyl groups are pointed in the same direction ($\theta = 0^\circ$), they eclipse each other. This conformation is called **totally eclipsed**, to distinguish it from the other eclipsed conformations, such as the one at $\theta = 120^\circ$. At $\theta = 60^\circ$, the butane molecule is staggered and the methyl groups are toward the left and right of each other. This 60° conformation is called **gauche** (pronounced gōsh), a French word meaning “left” or “awkward.”

Another staggered conformation occurs at $\theta = 180^\circ$, with the methyl groups pointing in opposite directions. This conformation is called **anti** because the methyl groups are “opposed.”

3-8A Torsional Energy of Butane

A graph of the relative torsional energies of the butane conformations is shown in Figure 3-11. All the staggered conformations (anti and gauche) are lower in energy than any of the eclipsed conformations. The anti conformation is lowest in energy because it places the bulky methyl groups as far apart as possible. The gauche conformations, with the methyl groups separated by just 60° , are 3.8 kJ (0.9 kcal) higher in energy than the anti conformation because the methyl groups are close enough that their electron clouds begin to repel each other. Use your molecular models to compare the crowding of the methyl groups in these conformations.

3-8B Steric Strain

The totally eclipsed conformation is about 6 kJ (1.4 kcal) higher in energy than the other eclipsed conformations because it forces the two end methyl groups so close together that their electron clouds experience a strong repulsion. This kind of

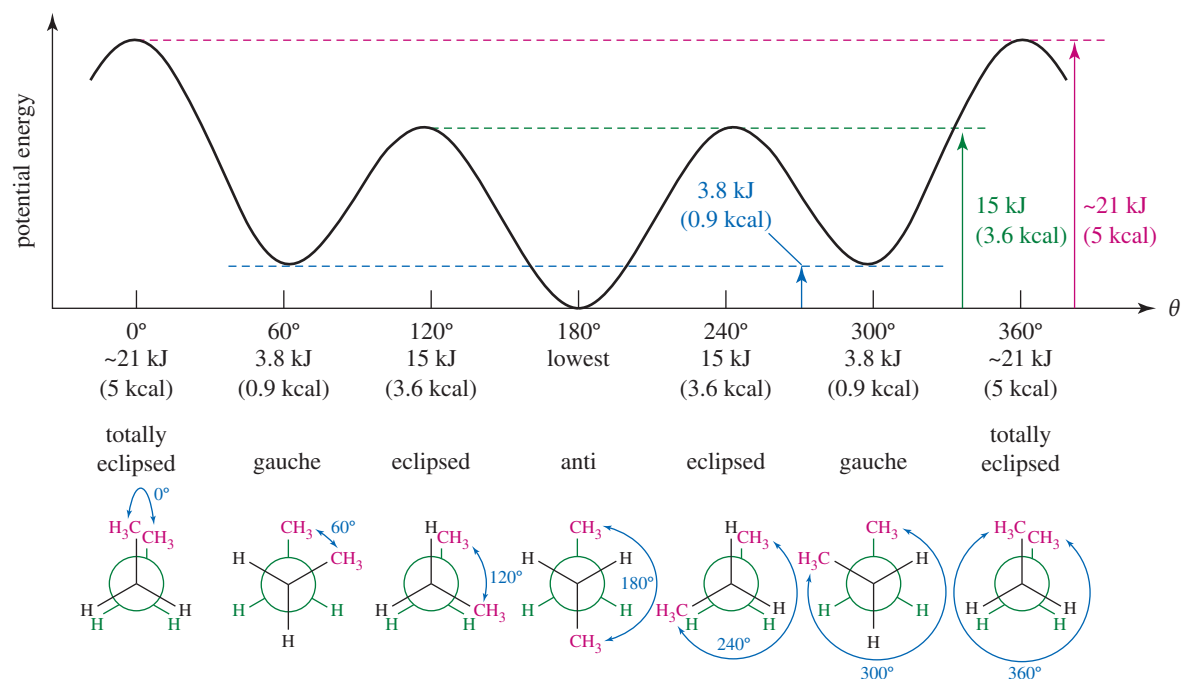
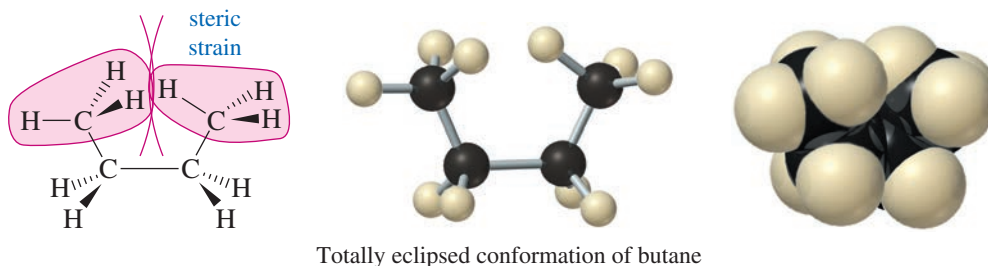


FIGURE 3-11 Torsional energy of butane. The anti conformation is lowest in energy, and the totally eclipsed conformation is highest in energy.

interference between two bulky groups is called **steric strain**.^{*} The following structure shows the interference between the methyl groups in the totally eclipsed conformation.



Rotating the totally eclipsed conformation 60° to a gauche conformation releases most, but not all, of this steric strain. The gauche conformation is still 3.8 kJ (0.9 kcal) higher in energy than the most stable anti conformation.

What we have learned about the conformations of butane can be applied to other alkanes. We can predict that carbon–carbon single bonds will assume staggered conformations whenever possible to avoid eclipsing the groups attached to them. Among the staggered conformations, the anti conformation is preferred because it has the lowest torsional energy. We must remember, however, that enough thermal energy is present at room temperature for the molecules to rotate rapidly among all the different conformations. The relative stabilities are important because more molecules will be found in the more stable conformations than in the less stable ones.

PROBLEM-SOLVING HINT

A C—CH₃ bond eclipsed with another C—CH₃ bond contributes about 13 kJ/mol (3 kcal/mol) torsional energy.

Bond	Eclipsed with	Molar energy
C—H	C—H	4.2 kJ (1.0 kcal)
C—H	C—CH ₃	5.4 kJ (1.3 kcal)
C—CH ₃	C—CH ₃	13 kJ (3 kcal)

PROBLEM 3-13

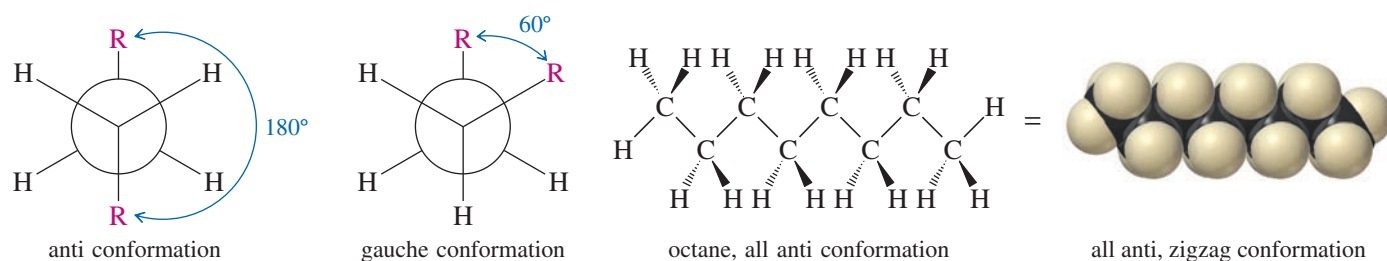
Draw a graph, similar to Figure 3-11, of the torsional energy of 2-methylbutane as it rotates about the C2—C3 bond.

^{*}“Steric strain” is sometimes called “steric hindrance,” a term that more appropriately refers to the slowing (hindrance) of a reaction because bulky groups interfere.

3-9 Conformations of Higher Alkanes

The higher alkanes resemble butane in their preference for anti and gauche conformations about the carbon–carbon bonds. The lowest-energy conformation for any straight-chain alkane is the one with all the internal carbon–carbon bonds in their anti conformations. These anti conformations give the chain a zigzag shape. At room temperature, the internal carbon–carbon bonds undergo rotation, and many molecules contain gauche conformations. Gauche conformations make kinks in the zigzag structure. Nevertheless, we frequently draw alkane chains in a zigzag structure to represent the most stable arrangement.

Conformations of long-chain alkyl groups are important in the chemistry of lipids (Chapter 25), with interesting implications for their melting points and for nutrition.



PROBLEM 3-14

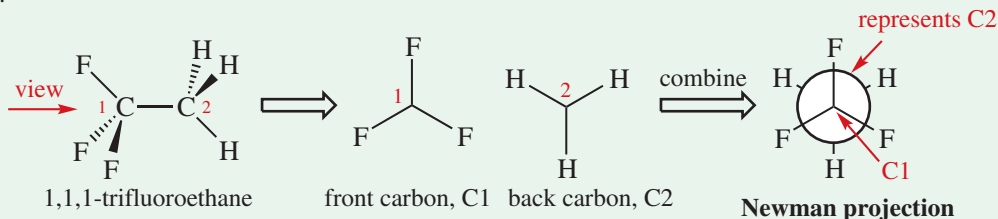
Draw a perspective representation of the most stable conformation of 3-methylhexane.

FOCUS Conformations of Acyclic Alkanes

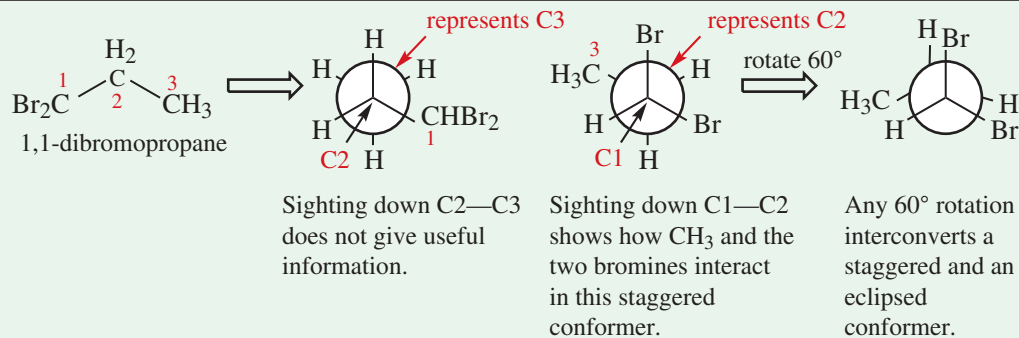
Atoms are continuously rotating around single bonds unless constrained. These different rotational positions are called **conformers**, short for *conformational isomers*, also known as *rotational isomers*. In theory, an infinite number of conformers can occur around a particular single bond, so to simplify, we study those at energy minima or maxima.

A convenient symbolism for conformers was devised by Prof. M.

Newman of The Ohio State University. The viewer looks straight down a C—C bond, and draws the groups as if they were coming out of a circle, with the bonds to C1 in front of the circle and bonds to C2 behind the circle.



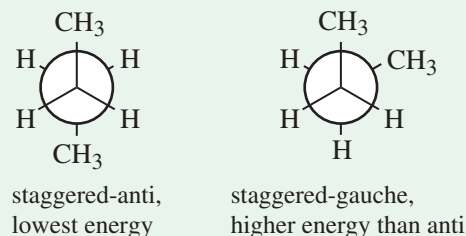
You can draw a Newman projection of any C—C bond, but some are more useful than others. If the C—C bond is not specified, choose one with the most groups other than H coming off the carbons. For example, consider 1,1-dibromopropane:



The lower-energy **staggered** conformer has groups as far from each other as possible, and the higher-energy **eclipsed** conformer has groups as close together as possible. In general, *all staggered conformers are lower in energy than any eclipsed conformer*. Depending on the groups on each carbon, there may be more than one staggered or eclipsed conformer, as shown directly below.

Any 60° rotation of a carbon in a Newman projection converts a staggered to an eclipsed conformer, or vice versa.

There are two types of staggered conformers, depending on the particular alkane represented. This Newman projection of butane, sighting down the C2—C3 bond, shows that C1 and C4 can be opposite each other in the **staggered-anti** conformer, or closer to one another in the **staggered-gauche** conformer. In most cases, a staggered-anti conformer has the lowest possible energy.



3-10 Cycloalkanes

Many organic compounds are **cyclic**: they contain rings of atoms. The carbohydrates we eat are cyclic, the nucleotides that make up our DNA and RNA are cyclic, and the antibiotics we use to treat diseases are cyclic. In this chapter, we use the *cycloalkanes* to illustrate the properties and stability of cyclic compounds.

Cycloalkanes are alkanes that contain rings of carbon atoms. Simple cycloalkanes are named like acyclic (noncyclic) alkanes, with the prefix *cyclo-* indicating the presence of a ring. For example, the cycloalkane with four carbon atoms in a ring is called *cyclobutane*. The cycloalkane with seven carbon atoms in a ring is *cycloheptane*. Line-angle formulas are often used for drawing the rings of cycloalkanes (Figure 3-12).

Application: Anesthesia

Cyclopropane was once used as general anesthetic because its vapors, like those of other simple alkanes and cycloalkanes, cause sleepiness and a loss of consciousness. After inhalation into the lungs, cyclopropane goes into the blood. Due to its nonpolar nature, it rapidly leaves the blood and passes through the nonpolar membranes surrounding the central nervous system, where it produces anesthesia. Cyclopropane is no longer used as an anesthetic because it is highly flammable (like ether) and can cause explosions when mixed with air.

3-10A General Molecular Formulas of Cycloalkanes

Simple cycloalkanes are rings of CH₂ groups (methylene groups). Each one has exactly twice as many hydrogen atoms as carbon atoms, giving the general molecular formula C_nH_{2n}. This general formula has two fewer hydrogen atoms than the (2n + 2) formula for an acyclic alkane because a ring has no ends, and no hydrogens are needed to cap off the ends of the chain.

3-10B Physical Properties of Cycloalkanes

Most cycloalkanes resemble the **acyclic** (noncyclic), open-chain alkanes in their physical properties and in their chemistry. They are nonpolar, relatively inert compounds with boiling points and melting points that depend on their molecular weights. The cycloalkanes are held in a more compact cyclic shape, so their physical properties are similar to those of the compact, branched alkanes. The physical properties of some common cycloalkanes are listed in Table 3-4.

TABLE 3-4 Physical Properties of Some Simple Cycloalkanes

Cycloalkane	Formula	Boiling Point (°C)	Melting Point (°C)	Density
cyclopropane	C ₃ H ₆	-33	-128	0.72
cyclobutane	C ₄ H ₈	-12	-50	0.75
cyclopentane	C ₅ H ₁₀	49	-94	0.75
cyclohexane	C ₆ H ₁₂	81	7	0.78
cycloheptane	C ₇ H ₁₄	118	-12	0.81
cyclooctane	C ₈ H ₁₆	148	14	0.83

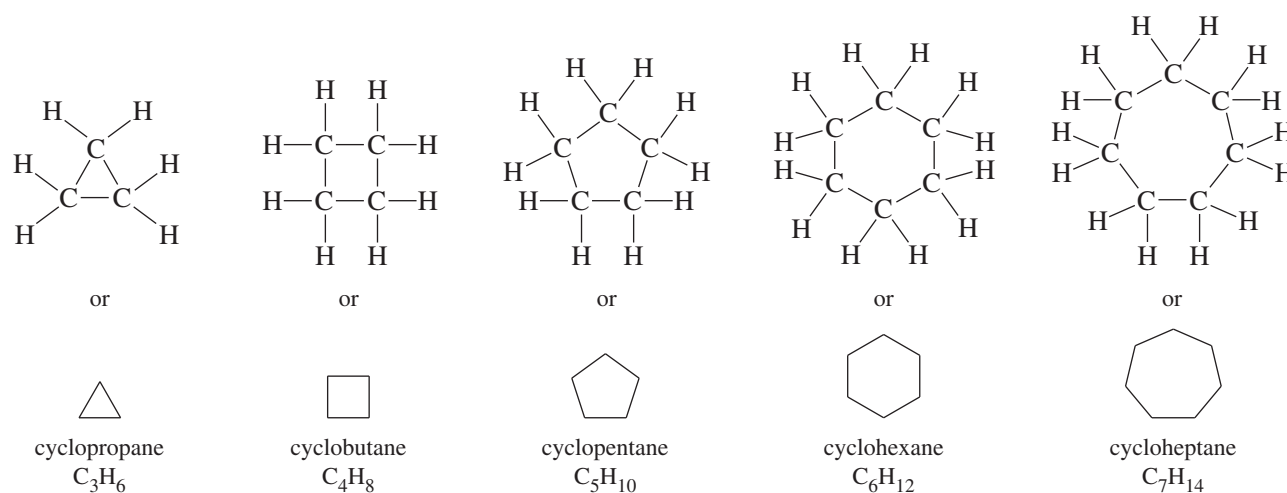
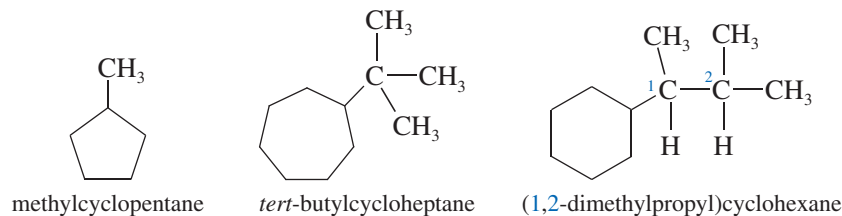


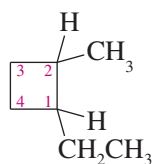
FIGURE 3-12 Structures of some cycloalkanes.

3-10C Nomenclature of Cycloalkanes

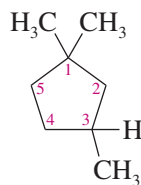
Cycloalkanes are named much like acyclic alkanes. Substituted cycloalkanes use the cycloalkane for the base name, with the alkyl groups named as substituents. If there is just one substituent, no numbering is needed.



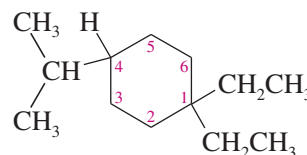
If there are two or more substituents on the ring, the ring carbons are numbered to give the lowest possible numbers for the substituted carbons. The numbering begins with one of the substituted ring carbons and continues in the direction that gives the lowest possible numbers to the other substituents. In the name, the substituents are listed in alphabetical order. When the numbering could begin with either of two substituted ring carbons (as in a disubstituted cycloalkane), begin with the one that has more substituents; otherwise, begin with the one that is first alphabetically.



1-ethyl-2-methylcyclobutane

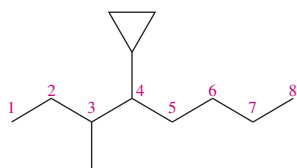


1,1,3-trimethylcyclopentane



1,1-diethyl-4-isopropylcyclohexane

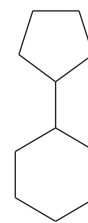
When the acyclic portion of the molecule contains more carbon atoms than the cyclic portion (or when it contains an important functional group), the cyclic portion is sometimes named as a cycloalkyl substituent.



4-cyclopropyl-3-methyloctane



1-cyclobutyl-3,5-dimethylhexane



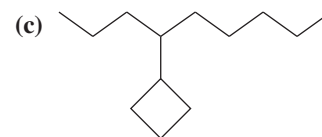
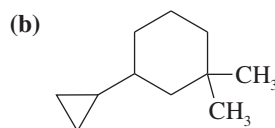
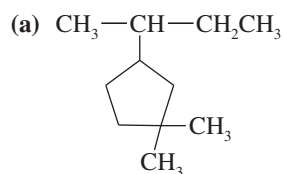
cyclopentylcyclohexane

PROBLEM-SOLVING HINT

Students accidentally draw cyclic structures when acyclic structures are intended and vice versa. Always verify whether the name contains the prefix **cyclo-**.

PROBLEM 3-15

Give IUPAC names for the following compounds.

**PROBLEM 3-16**

Draw the structure and give the molecular formula for each of the following compounds.

(a) 1-ethyl-3-methylcycloheptane

(b) isobutylcyclohexane

(c) cyclopropylcyclopentane

(d) 3-ethyl-1,1-dimethylcyclohexane

(e) 3-ethyl-2,4-dimethylhexane

(f) 1,1-diethyl-4-(3,3-dimethylbutyl)cyclohexane

3-11 Cis-trans Isomerism in Cycloalkanes

Open-chain (acyclic) alkanes undergo rotations about their carbon–carbon single bonds, so they are free to assume any of an infinite number of conformations. Alkenes have rigid double bonds that prevent rotation, giving rise to *cis* and *trans* isomers with different orientations of the groups on the double bond (Section 1-19B). Cycloalkanes are similar to alkenes in this respect. A cycloalkane has two distinct faces. If two substituents point toward the same face, they are **cis**. If they point toward opposite faces, they are **trans**. These **geometric isomers** cannot interconvert without breaking and re-forming bonds.

Figure 3-13 compares the *cis-trans* isomers of but-2-ene with those of 1,2-dimethylcyclopentane. Make models of these compounds to convince yourself that *cis*- and *trans*-1,2-dimethylcyclopentane cannot interconvert by simple rotations about the bonds. Note that in the perspective drawings, the thicker lines indicate the parts of the molecule that extend out toward the viewer.

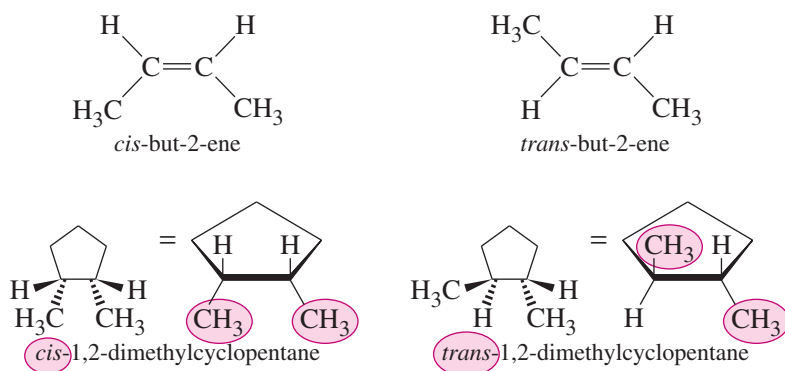


FIGURE 3-13 Cis-trans isomerism in cycloalkanes. Like alkenes, cycloalkane rings are restricted from free rotation. Two substituents on a cycloalkane must be on the same side (*cis*) or on opposite sides (*trans*) of the ring.

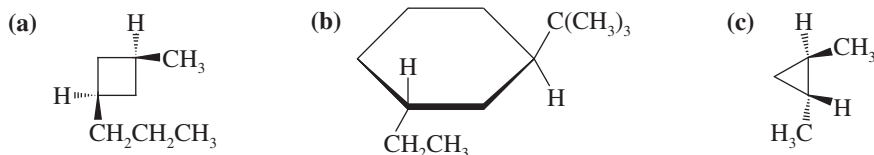
PROBLEM 3-17

Which of the following cycloalkanes are capable of geometric (*cis-trans*) isomerism? Draw the *cis* and *trans* isomers.

- (a) 3-ethyl-1,1-dimethylcyclohexane (b) 1-ethyl-3-methylcycloheptane
(c) 1-ethyl-3-methylcyclopentane (d) 1-cyclopropyl-2-methylcyclohexane

PROBLEM 3-18

Give IUPAC names for the following cycloalkanes.



3-12 Stabilities of Cycloalkanes; Ring Strain

Although all the simple cycloalkanes (up to about C_{20}) have been synthesized, the most common rings contain five or six carbon atoms. We will study the stabilities and conformations of these rings in detail because they help to determine the properties of many important organic compounds.

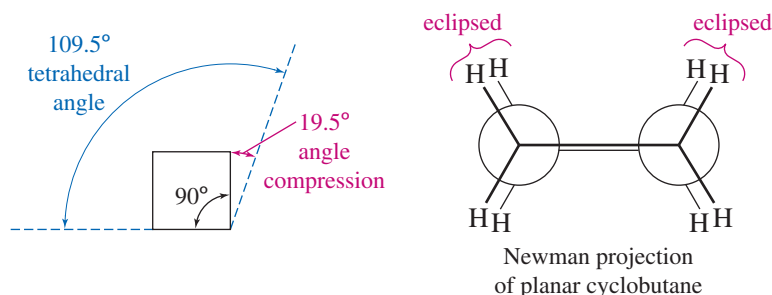
Why are five-membered and six-membered rings more common than the other sizes? Adolf von Baeyer first attempted to explain the relative stabilities of cyclic molecules in the late 19th century, and he was awarded a Nobel Prize for this work in 1905. Baeyer reasoned that the carbon atoms in acyclic alkanes have bond angles of 109.5° . (We now explain this bond angle by the tetrahedral geometry of the sp^3 hybridized carbon atoms.)

If a cycloalkane requires bond angles other than 109.5° , the orbitals of its carbon-carbon bonds cannot achieve optimum overlap, and the cycloalkane must have some **angle strain** (sometimes called **Baeyer strain**) associated with it. Figure 3-14 shows that a planar cyclobutane, with 90° bond angles, is expected to have significant angle strain.

In addition to this angle strain, the Newman projection in Figure 3-14 shows that the bonds are eclipsed, resembling the *totally eclipsed* conformation of butane (Section 3-7). This eclipsing of bonds gives rise to torsional strain. Together, the angle strain and the torsional strain add to give what we call the **ring strain** of the cyclic compound. The amount of ring strain depends primarily on the size of the ring.

Before we discuss the ring strain of different cycloalkanes, we need to consider how ring strain is measured. In theory, we should measure the total amount of energy in the cyclic compound and subtract the amount of energy in a similar strain-free reference

FIGURE 3-14 The ring strain of a planar cyclobutane results from two factors: angle strain from the compressing of the bond angles to 90° rather than the tetrahedral angle of 109.5° , and torsional strain from eclipsing of the bonds.

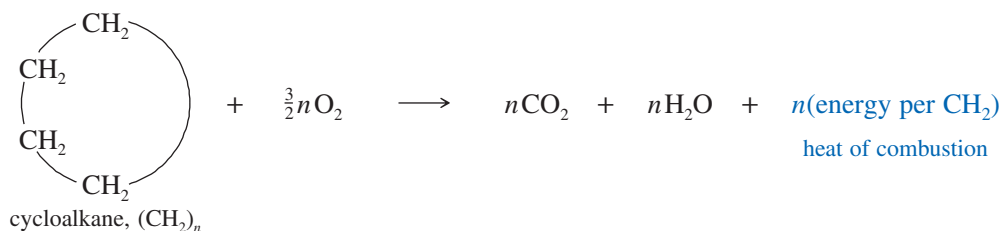


compound. The difference should be the amount of extra energy due to ring strain in the cyclic compound. These measurements are commonly made using *heats of combustion*.

3-12A Heats of Combustion

The **heat of combustion** is the amount of heat released when a compound is burned with an excess of oxygen in a sealed container called a *bomb calorimeter*. If the compound has extra energy as a result of ring strain, that extra energy is released in the combustion. The heat of combustion is usually measured by the temperature rise in the water bath surrounding the “bomb.”

A cycloalkane can be represented by the molecular formula $(\text{CH}_2)_n$, so the general reaction in the bomb calorimeter is



The molar heat of combustion of cyclohexane is nearly twice that of cyclopropane, simply because cyclohexane contains twice as many methylene (CH_2) groups per mole. To compare the relative stabilities of cycloalkanes, we divide the heat of combustion by the number of methylene (CH_2) groups. The result is the energy per CH_2 group. These normalized energies allow us to compare the relative amounts of ring strain (per methylene group) in the cycloalkanes.

Table 3-5 lists the heats of combustion for some simple cycloalkanes. The reference value of 658.6 kJ (157.4 kcal) per mole of CH_2 groups comes from an unstrained long-chain alkane. The values show large amounts of ring strain in cyclopropane and cyclobutane. Cyclopentane, cycloheptane, and cyclooctane have much smaller amounts of ring strain, and cyclohexane has no ring strain at all. We will discuss several of these rings in detail to explain this pattern of ring strain.

3-12B Cyclopropane

Table 3-5 shows that cyclopropane bears more ring strain per methylene group than any other cycloalkane. Two factors contribute to this large ring strain. First is the angle strain required to compress the bond angles from the tetrahedral angle of 109.5° to the 60° angles of cyclopropane. The bonding overlap of the carbon–carbon sp^3 orbitals is weakened when the bond angles differ so much from the tetrahedral angle. The sp^3 orbitals cannot point directly toward each other, and they overlap at an angle to form weaker “bent bonds” (Figure 3-15).

Torsional strain is the second factor in cyclopropane’s large ring strain. The three-membered ring is planar, and all the bonds are eclipsed. A Newman projection of one of the carbon–carbon bonds (Figure 3-16) shows that the conformation resembles the

TABLE 3-5 Heats of Combustion (per Mole) for Some Simple Cycloalkanes

Ring Size	Cycloalkane	Molar Heat of Combustion	Heat of Combustion per CH ₂ Group	Ring Strain per CH ₂ Group	Total Ring Strain
3	cyclopropane	2091 kJ (499.8 kcal)	697.1 kJ (166.6 kcal)	38.5 kJ (9.2 kcal)	115 kJ (27.6 kcal)
4	cyclobutane	2744 kJ (655.9 kcal)	686.1 kJ (164.0 kcal)	27.5 kJ (6.6 kcal)	110 kJ (26.3 kcal)
5	cyclopentane	3320 kJ (793.5 kcal)	664.0 kJ (158.7 kcal)	5.4 kJ (1.3 kcal)	27 kJ (6.5 kcal)
6	cyclohexane	3951 kJ (944.4 kcal)	658.6 kJ (157.4 kcal)	0.0 kJ (0.0 kcal)	0.0 kJ (0.0 kcal)
7	cycloheptane	4637 kJ (1108.2 kcal)	662.4 kJ (158.3 kcal)	3.8 kJ (0.9 kcal)	27 kJ (6.4 kcal)
8	cyclooctane	5309 kJ (1268.9 kcal)	663.6 kJ (158.6 kcal)	5.1 kJ (1.2 kcal)	41 kJ (9.7 kcal)
reference: long-chain alkane			658.6 kJ (157.4 kcal)	0.0 kJ (0.0 kcal)	0.0 kJ (0.0 kcal)

All units are per mole.

totally eclipsed conformation of butane. The torsional strain in cyclopropane is not as great as its angle strain, but it helps to account for the large total ring strain.

Cyclopropane is generally more reactive than other alkanes. Reactions that open the cyclopropane ring release 115 kJ (27.6 kcal) per mole of ring strain, which provides an additional driving force for these reactions.

PROBLEM 3-19

The heat of combustion of *cis*-1,2-dimethylcyclopropane is larger than that of the *trans* isomer. Which isomer is more stable? Use drawings to explain this difference in stability.

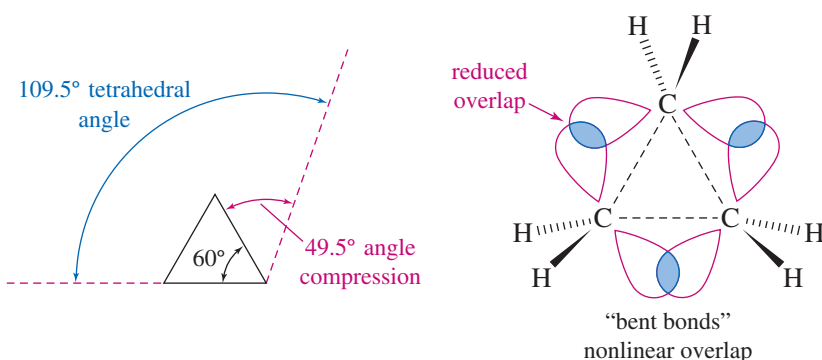


FIGURE 3-15 Angle strain in cyclopropane. The bond angles are compressed to 60° from the usual 109.5° bond angle of sp^3 hybridized carbon atoms. This severe angle strain leads to nonlinear overlap of the sp^3 orbitals and “bent bonds.”

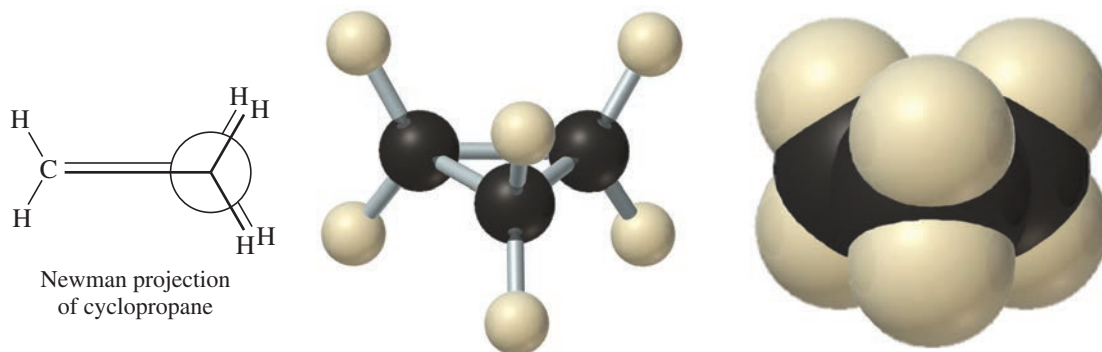


FIGURE 3-16 Torsional strain in cyclopropane. All the carbon–carbon bonds are eclipsed, generating torsional strain that contributes to the total ring strain.

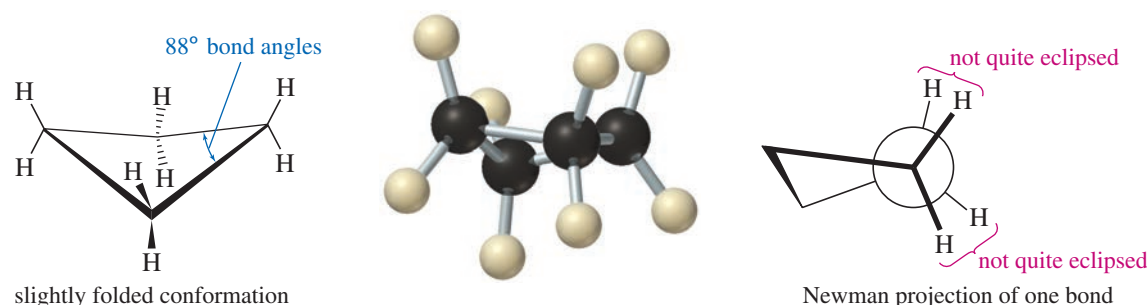


FIGURE 3-17 The conformation of cyclobutane is slightly folded. Folding gives partial relief from the eclipsing of bonds, as shown in the Newman projection. Compare this actual structure with the hypothetical planar structure in Figure 3-14.

3-12C Cyclobutane

The total ring strain in cyclobutane is almost as great as that in cyclopropane, but is distributed over four carbon atoms. If cyclobutane were perfectly planar and square, it would have 90° bond angles. A planar geometry requires eclipsing of all the bonds, however, as in cyclopropane. To reduce this torsional strain, cyclobutane actually assumes a slightly folded form, with bond angles of 88°. These smaller bond angles require slightly more angle strain than 90° angles, but the relief of some of the torsional strain appears to compensate for a small increase in angle strain (Figure 3-17).

PROBLEM 3-20

trans-1,2-Dimethylcyclobutane is more stable than *cis*-1,2-dimethylcyclobutane, but *cis*-1,3-dimethylcyclobutane is more stable than *trans*-1,3-dimethylcyclobutane. Use drawings to explain these observations.

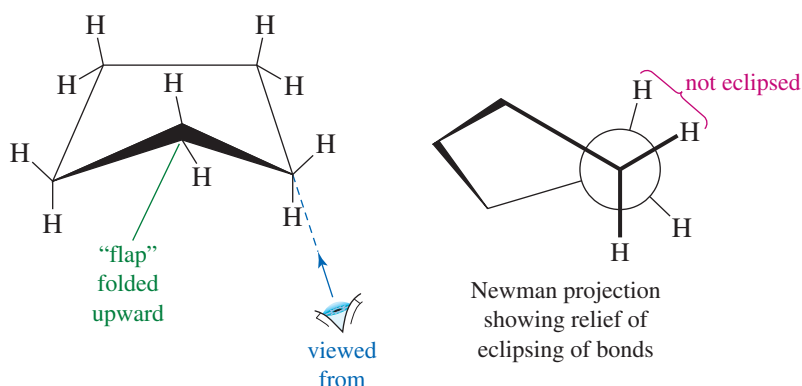
Application: Biochemistry

The conformation of cyclopentane is important because ribose and deoxyribose, the sugar components of RNA and DNA, respectively, assume cyclopentane-like ring conformations. These conformations are crucial to the properties and reactions of RNA and DNA.

3-12D Cyclopentane

If cyclopentane had the shape of a planar, regular pentagon, its bond angles would be 108°, close to the tetrahedral angle of 109.5°. A planar structure would require all the bonds to be eclipsed, however. Cyclopentane actually assumes a slightly puckered “envelope” conformation that reduces the eclipsing and lowers the torsional strain (Figure 3-18). This puckered shape is not fixed, but undulates by the thermal up-and-down motion of the five methylene groups. The “flap” of the envelope seems to move around the ring as the molecule undulates.

FIGURE 3-18 The conformation of cyclopentane is slightly folded, like the shape of an envelope. This puckered conformation reduces the eclipsing of adjacent CH₂ groups, which reduces the torsional strain.



3-13 Cyclohexane Conformations

We will cover the conformations of cyclohexane in more detail than other cycloalkanes because cyclohexane ring systems are particularly common. Carbohydrates, steroids, plant products, pesticides, and many other important compounds contain cyclohexane-like rings whose conformations and stereochemistry are critically important to their reactivity. The abundance of cyclohexane rings in nature is probably due to both their stability and the selectivity offered by their predictable conformations. Nature probably forms more six-membered rings than all other ring sizes combined.

The combustion data (Table 3-5) show that cyclohexane has *no* ring strain. Cyclohexane must have bond angles that are near the tetrahedral angle (no angle strain) and also have no eclipsing of bonds (no torsional strain). A planar, regular hexagon would have bond angles of 120° rather than 109.5° , implying some angle strain. A planar ring would also have torsional strain because the bonds on adjacent CH_2 groups would be eclipsed. Therefore, the cyclohexane ring cannot be planar.

3-13A Chair and Boat Conformations

Cyclohexane achieves tetrahedral bond angles and staggered conformations by assuming a puckered conformation. The most stable conformation is the **chair conformation** shown in Figure 3-19. Build a molecular model of cyclohexane, and compare its shape with the drawings in Figure 3-19. In the chair conformation, the angles between the carbon–carbon bonds are all 109.5° . The Newman projection looking down the “seat” bonds shows all the bonds in staggered conformations.

The **boat conformation** of cyclohexane (Figure 3-20) also has bond angles of 109.5° and avoids angle strain. The boat conformation resembles the chair conformation except that the “footrest” methylene group is folded upward. The boat conformation suffers from torsional strain, however, because there is eclipsing of bonds.

This eclipsing forces two of the hydrogens on the ends of the “boat” to interfere with each other. These hydrogens are called **flagpole hydrogens** because they point upward from the ends of the boat like two flagpoles. The Newman projection in Figure 3-20 shows this eclipsing of the carbon–carbon bonds along the sides of the boat.

A cyclohexane molecule in the boat conformation actually exists as a slightly skewed **twist boat conformation**, also shown in Figure 3-20. If you assemble your molecular model in the boat conformation and twist it slightly, the flagpole hydrogens move away from each other and the eclipsing of the bonds is reduced. Even though the twist boat is lower in energy than the symmetrical boat, it is still about 23 kJ/mol (5.5 kcal/mol) higher in energy than the chair conformation. When someone refers to the “boat conformation,” the twist boat (or simply twist) conformation is often intended.

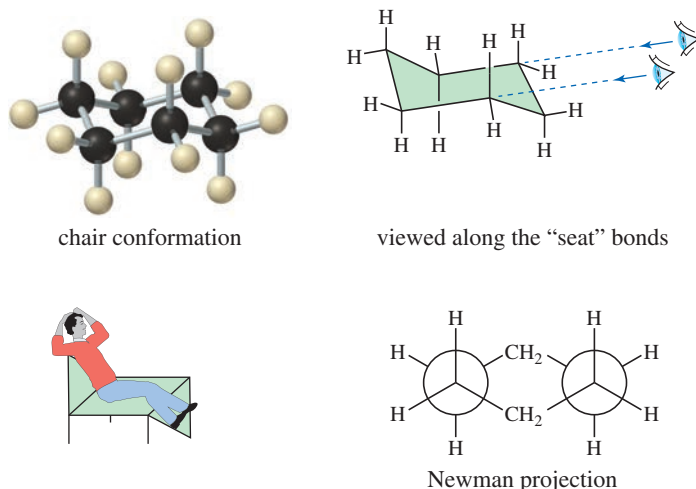
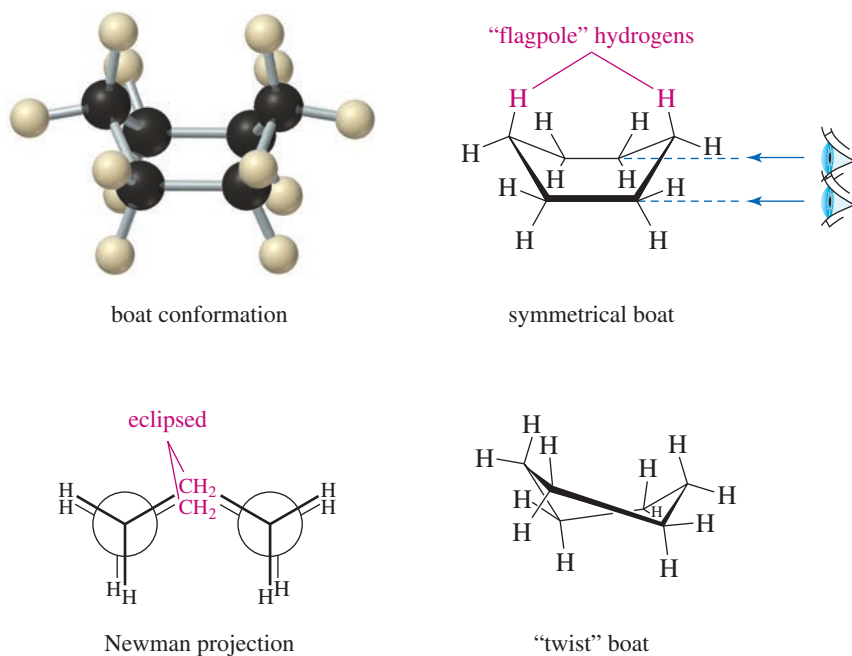


FIGURE 3-19 Viewed from the side, the chair conformation of cyclohexane appears to have one methylene group puckered upward and another puckered downward. Viewed from the Newman projection, the chair has no eclipsing of the carbon–carbon bonds. The bond angles are 109.5° .

FIGURE 3-20 In the symmetrical boat conformation of cyclohexane, eclipsing of bonds results in torsional strain. In the actual molecule, the boat is skewed to give the twist boat, a conformation with less eclipsing of bonds and less interference between the two flagpole hydrogens.



Application: Biochemistry

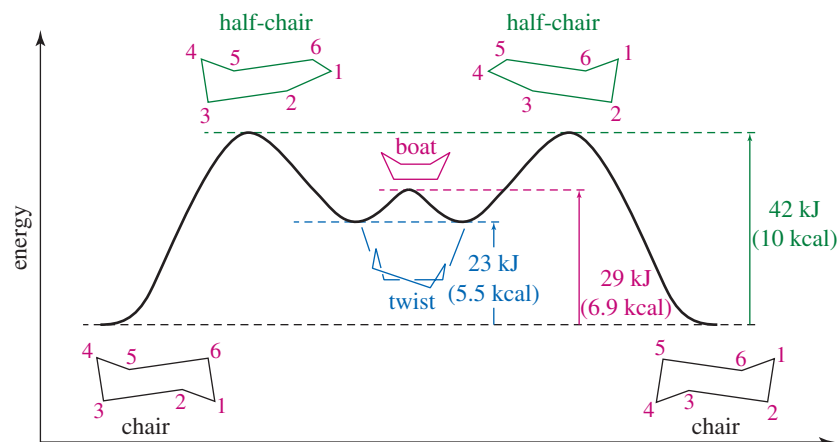
The conformations of biological molecules are critical for their activities. For example, steroids fit into their receptors in only one conformation. The correct fit activates the receptor, resulting in a biological response.

At any instant, most of the molecules in a cyclohexane sample are in chair conformations. The energy barrier between the boat and chair is sufficiently low, however, that the conformations interconvert many times each second. The interconversion from the chair to the boat takes place by the footrest of the chair flipping upward and forming the boat. The highest-energy point in this process is the conformation where the footrest is planar with the sides of the molecule. This unstable arrangement is called the **half-chair conformation**. Figure 3-21 shows how the energy of cyclohexane varies as it interconverts between the boat and chair forms.

3-13B Axial and Equatorial Positions

If we could freeze cyclohexane in a chair conformation, we would see that there are two different orientations of carbon–hydrogen bonds. Six of the bonds (one on each carbon atom) are directed up and down, parallel to the axis of the ring. These are called **axial bonds**. The other six bonds point out from the ring, along the “equator” of the ring. These are called **equatorial bonds**. The axial bonds and hydrogens are shown in red in Figure 3-22, and the equatorial bonds and hydrogens are shown in green.

FIGURE 3-21 Conformational energy of cyclohexane. The chair conformation is most stable, followed by the twist boat. To convert between these two conformations, the molecule must pass through the unstable half-chair conformation.



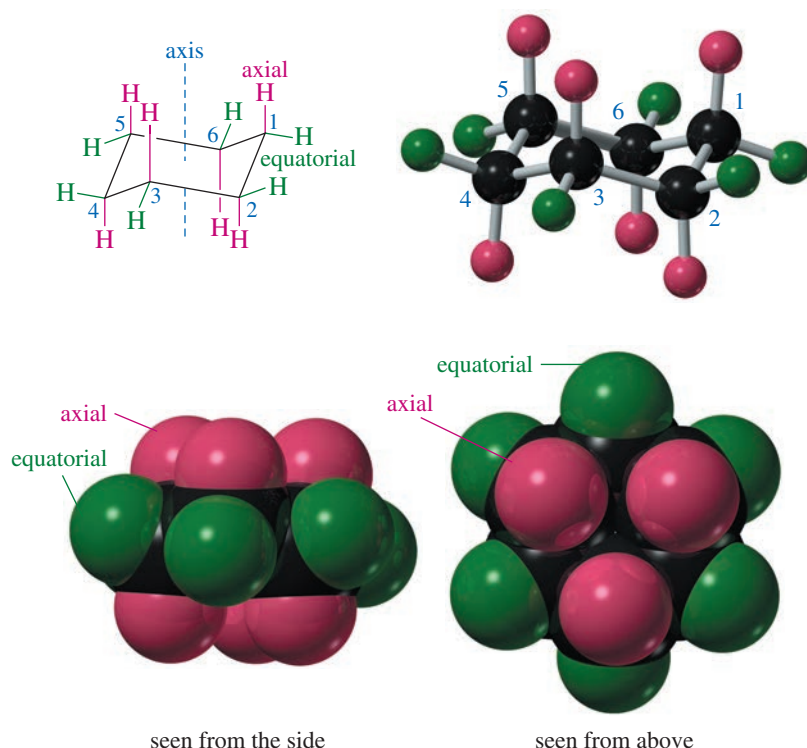


FIGURE 3-22 Axial bonds are directed vertically, parallel to the axis of the ring. Equatorial bonds are directed outward, from the equator of the ring. As they are numbered here, the odd-numbered carbons have their *upward* bonds axial and their *downward* bonds equatorial. The even-numbered carbons have their *downward* bonds axial and their *upward* bonds equatorial.

Each carbon atom in cyclohexane is bonded to two hydrogen atoms, one directed upward and one downward. As the carbon atoms are numbered in Figure 3-22, C1 has an axial bond upward and an equatorial bond downward. C2 has an equatorial bond upward and an axial bond downward. The pattern alternates. The odd-numbered carbon atoms have axial bonds up and equatorial bonds down, like C1. The even-numbered carbons have equatorial bonds up and axial bonds down, like C2. This pattern of alternating axial and equatorial bonds is helpful for predicting the conformations of substituted cyclohexanes, as we see in Sections 3-13 and 3-14.

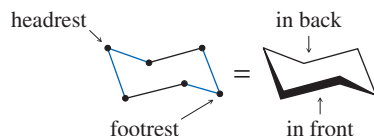
PROBLEM-SOLVING STRATEGY Drawing Chair Conformations

Drawing realistic pictures of cyclohexane conformations is not difficult, but certain rules should be followed to show the actual positions and angles of the substituents on the ring. Make a cyclohexane ring with your models, put it in a chair conformation, and use it to follow along with this discussion. When you hold your model at the angle that corresponds to a drawing, the angles of the bonds in the model should correspond to the angles in the drawing.

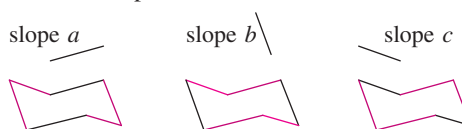
To draw the carbon-carbon bond framework, first draw two parallel lines slightly slanted and slightly offset. The atoms at the ends of these bonds lie in a plane, and they define what will be the “armrests” of our chair.



Draw the headrest and footrest carbons, and draw the lines connecting them to the armrests. The two lines connecting the headrest carbon should be parallel to the two lines connecting the footrest.



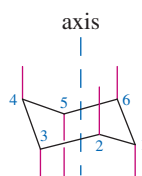
Notice that the carbon–carbon bond framework uses lines with only three different slopes, labeled *a*, *b*, and *c*. Compare this drawing with your model, and notice the pairs of carbon–carbon bonds with three distinct slopes.



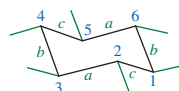
We can draw the chair with the headrest to the left and the footrest to the right or vice versa. Practice drawing it both ways.



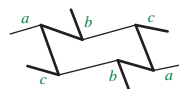
Now fill in the axial and equatorial bonds. The axial bonds are drawn vertically, either up or down. When a vertex of the chair points upward, its axial bond also points upward. If the vertex points downward, its axial bond points downward. C1 is a downward-pointing vertex, and its axial bond also points downward. C2 points upward, and its axial bond points upward.



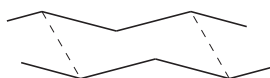
The equatorial bonds take more thought. Each carbon atom is represented by a vertex formed by two lines (bonds) having two of the possible slopes *a*, *b*, and *c*. Each equatorial bond should have the third slope: the slope that is *not* represented by the two lines forming the vertex.



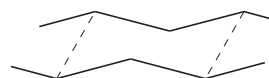
Look at your model as you add the equatorial bonds. The vertex C1 is formed by lines of slopes *b* and *c*, so its equatorial bond should have slope *a*. The equatorial bond at C2 should have slope *b*, and so on. Notice the W- and M-shaped patterns that result when these bonds are drawn correctly.



If your cyclohexane rings look awkward or slanted when using the analytical approach just shown, then try the artistic approach:* Draw a wide M, and draw a wide W below it, displaced about half a bond length to one side or the other. Connect the second atoms and the fourth atoms to give the cyclohexane ring with four equatorial bonds.

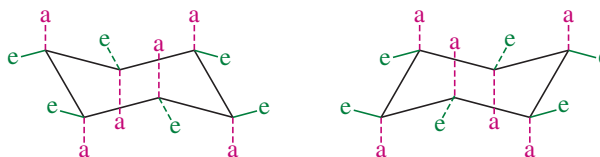


W displaced to the right



W displaced to the left

The other two equatorial bonds are drawn parallel to the ring connections. The axial bonds are then drawn vertically.



*See V. Dragojlovic, *J. Chem. Educ.* 2001, 78, 923.

PROBLEM 3-21

The cyclohexane chair shown in Figure 3-22 has the headrest to the right and the footrest to the left. Draw a cyclohexane chair with its axial and equatorial bonds, showing the headrest to the left and the footrest to the right.

PROBLEM 3-22

Draw 1,2,3,4,5,6-hexamethylcyclohexane with all the methyl groups
(a) in axial positions. (b) in equatorial positions.

3-14 Conformations of Monosubstituted Cyclohexanes

A substituent on a cyclohexane ring (in the chair conformation) can occupy either an axial or equatorial position. In many cases, the reactivity of the substituent depends on whether its position is axial or equatorial. The two possible chair conformations for methylcyclohexane are shown in Figure 3-23. These conformations are in equilibrium because they interconvert at room temperature. The boat (actually the twist boat) serves as an intermediate in this **chair–chair interconversion**, sometimes called a “ring-flip.” Place different-colored atoms in the axial and equatorial positions of your cyclohexane model, and notice that the chair–chair interconversion changes axial to equatorial and equatorial to axial.

The two chair conformations of methylcyclohexane interconvert at room temperature, so the one that is lower in energy predominates. Careful measurements have shown that the chair with the methyl group in an equatorial position is the most stable conformation. It is about 7.6 kJ/mol (1.8 kcal/mol) lower in energy than the conformation with the methyl group in an axial position. Both of these chair conformations are lower in energy than any boat conformation. We can show how the 7.6 kJ energy difference

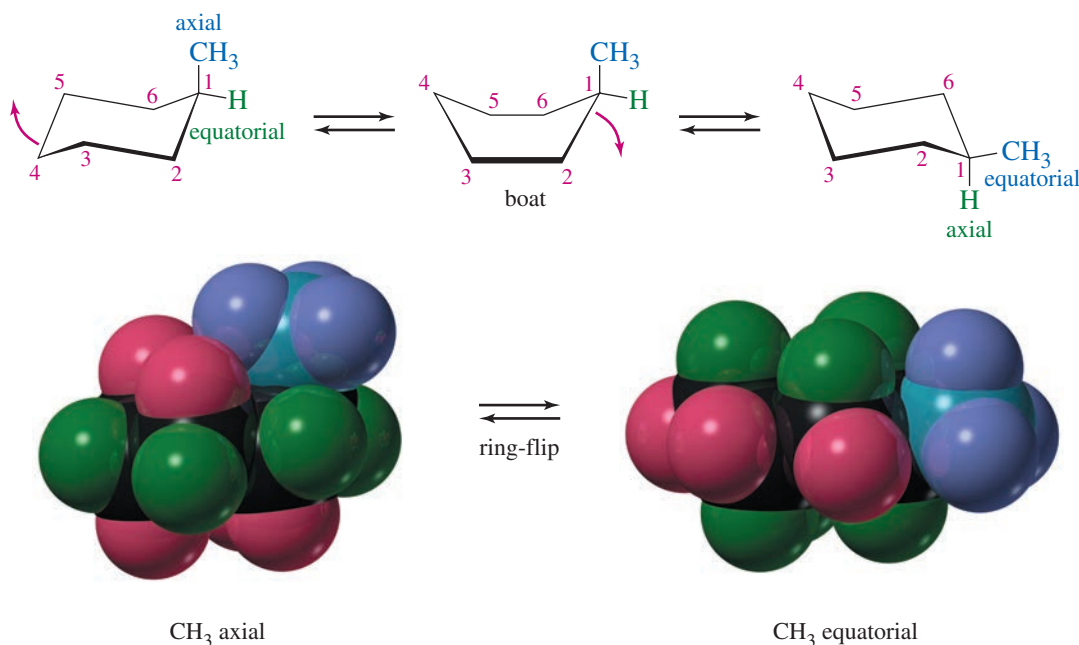
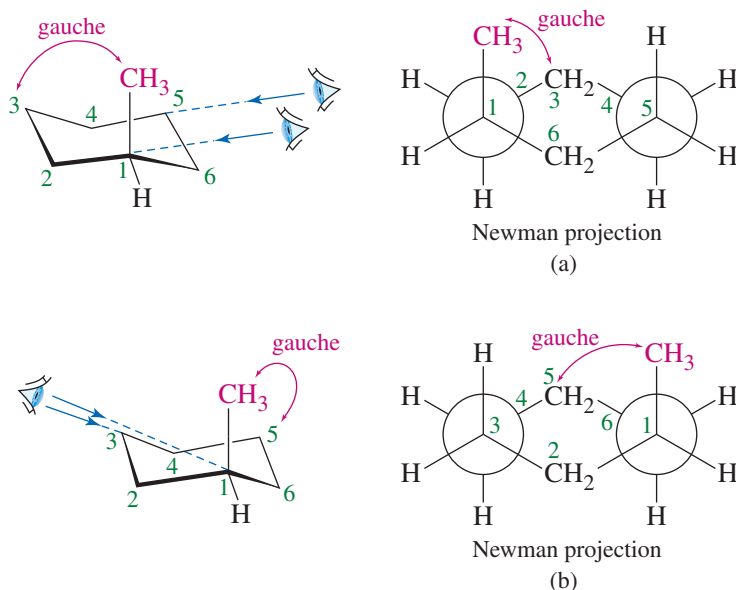


FIGURE 3-23 Chair–chair interconversion of methylcyclohexane. The methyl group is axial in one conformation, and equatorial in the other.

FIGURE 3-24 (a) When the methyl substituent is in an axial position on C1, it is gauche to C3. (b) The axial methyl group on C1 is also gauche to C5 of the ring.



between the axial and equatorial positions arises by examining molecular models and Newman projections of the two conformations. First, make a model of methylcyclohexane and use it to follow this discussion.

Consider a Newman projection looking along the armrest bonds of the conformation with the methyl group axial (Figure 3-24a): the methyl group is on C1, and we are looking from C1 toward C2. There is a 60° angle between the bond to the methyl group and the bond from C2 to C3, placing the axial methyl substituent and C3 in a gauche relationship. This axial methyl group is also gauche to C5, as you will see if you look along the C1—C6 bond in your model. Figure 3-24b shows this second gauche relationship.

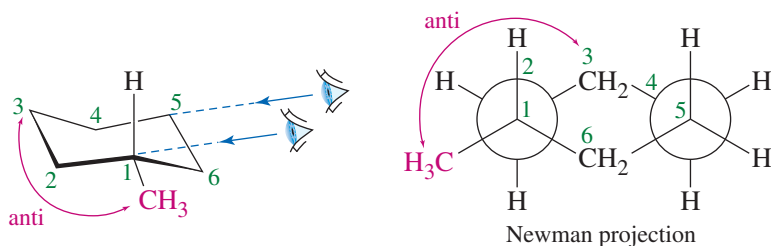
The Newman projection for the conformation with the methyl group equatorial shows that the methyl group has an anti relationship to both C3 and C5. Figure 3-25 shows the Newman projection along the C1—C2 bond, with the anti relationship of the methyl group to C3.

PROBLEM 3-23

Draw a Newman projection, similar to Figure 3-25, down the C1—C6 bond in the equatorial conformation of methylcyclohexane. Show that the equatorial methyl group is also anti to C5. (Using your models will help.)

The axial methylcyclohexane conformation has two gauche interactions, each representing about 3.8 kJ (0.9 kcal) of additional energy. The equatorial methyl group has no gauche interactions. Therefore, we predict that the axial conformation is higher

FIGURE 3-25 Looking down the C1—C2 bond of the equatorial conformation. Notice that the methyl group is anti to C3.



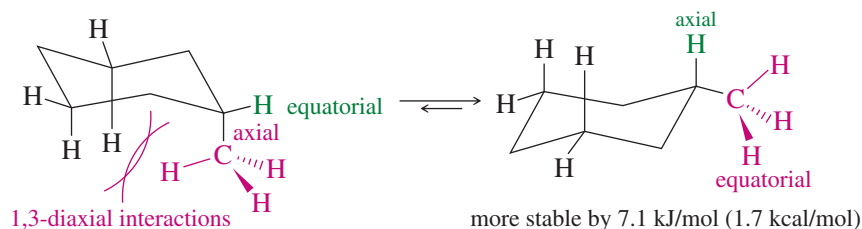


FIGURE 3-26 The axial substituent interferes with the axial hydrogens on C3 and C5. This interference is called a 1,3-diaxial interaction.

in energy by 7.6 kJ (1.8 kcal) per mole, in good agreement with the experimental value. Figure 3-26 shows that the gauche relationship of the axial methyl group with C3 and C5 places the methyl hydrogens close to the axial hydrogens on these carbons, causing their electron clouds to interfere. This form of steric strain is called a **1,3-diaxial interaction** because it involves substituents on carbon atoms of the ring that bear a 1,3 relationship. These 1,3-diaxial interactions are not present in the equatorial conformation.

A larger group usually has a larger energy difference between the axial and equatorial positions, because the 1,3-diaxial interaction shown in Figure 3-26 is stronger for larger groups. Table 3-6 lists the energy differences between the axial and equatorial positions for several alkyl groups and functional groups. The axial position is higher in energy in each case.

PROBLEM 3-24

Table 3-6 shows that the axial–equatorial energy difference for methyl, ethyl, and isopropyl groups increases gradually: 7.6, 7.9, and 8.8 kJ/mol (1.8, 1.9, and 2.1 kcal/mol). The *tert*-butyl group jumps to an energy difference of 23 kJ/mol (5.4 kcal/mol), over twice the value for the isopropyl group. Draw pictures of the axial conformations of isopropylcyclohexane and *tert*-butylcyclohexane, and explain why the *tert*-butyl substituent experiences such a large increase in axial energy over the isopropyl group.

PROBLEM 3-25

Draw the most stable conformation of

(a) ethylcyclohexane.

(b) 3-isopropyl-1,1-dimethylcyclohexane.

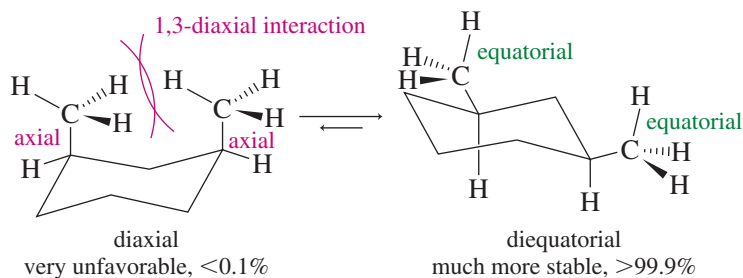
(c) *cis*-1-*tert*-butyl-4-isopropylcyclohexane.

TABLE 3-6 Energy Differences Between the Axial and Equatorial Conformations of Monosubstituted Cyclohexanes

	X	ΔG (axial–equatorial)	
		(kJ/mol)	(kcal/mol)
	—F	0.8	0.2
	—CN	0.8	0.2
	—Cl	2.1	0.5
	—Br	2.5	0.6
	—OH	4.1	1.0
	—COOH	5.9	1.4
	—CH ₃	7.6	1.8
	—CH ₂ CH ₃	7.9	1.9
	—CH(CH ₃) ₂	8.8	2.1
	—C(CH ₃) ₃	23	5.4

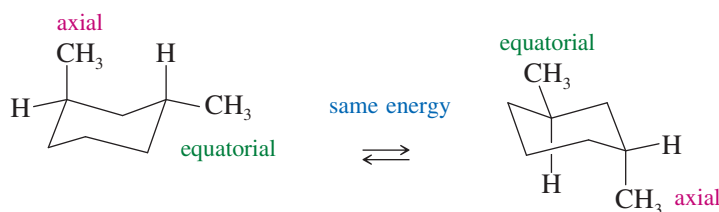
3-15 Conformations of Disubstituted Cyclohexanes

The steric interference between substituents in axial positions is particularly severe when there are large groups on two carbon atoms that bear a 1,3-diaxial relationship (*cis* on C1 and C3, or on C1 and C5), as in the two chair conformations of *cis*-1,3-dimethylcyclohexane shown here. The less stable conformation has both methyl groups in axial positions. The more stable conformation has both methyl groups in equatorial positions. Note the strongly unfavorable 1,3-diaxial interaction between the two methyl groups in the diaxial conformation. The molecule can relieve this 1,3-diaxial interference by flipping to the diequatorial conformation. Use your models to compare the diaxial and diequatorial forms of *cis*-1,3-dimethylcyclohexane.



trans-1,3-Dimethylcyclohexane does not have a conformation with a 1,3-diaxial interaction between two methyl groups. Either of its chair conformations places one methyl group in an axial position and one in an equatorial position. These conformations have equal energies, and they are present in equal amounts.

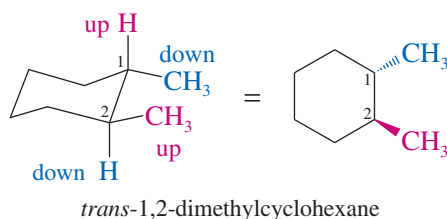
Chair conformations of *trans*-1,3-dimethylcyclohexane



Now we can compare the relative stabilities of the *cis* and *trans* isomers of 1,3-dimethylcyclohexane. The most stable conformation of the *cis* isomer has both methyl groups in equatorial positions. Either conformation of the *trans* isomer places one methyl group in an axial position. The *trans* isomer is therefore higher in energy than the *cis* isomer by about 7.6 kJ/mol (1.8 kcal/mol), the energy difference between axial and equatorial methyl groups. Remember that the *cis* and *trans* isomers cannot interconvert, and there is no equilibrium between these isomers.

PROBLEM-SOLVING STRATEGY Recognizing Cis and Trans Isomers

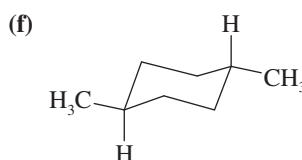
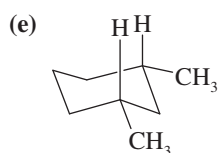
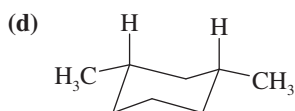
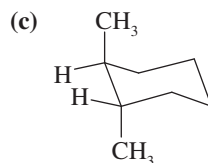
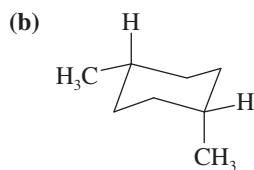
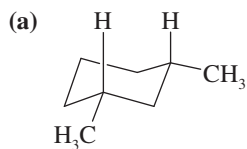
Some students find it difficult to look at a chair conformation and tell whether a disubstituted cyclohexane is the *cis* isomer or the *trans* isomer. In the following drawing, the two methyl groups appear to be oriented in similar directions. They are actually *trans* but are often mistaken for *cis*.



This ambiguity is resolved by recognizing that each of the ring carbons has two available bonds, one upward and one downward. In this drawing, the methyl group on C1 is on the downward bond, and the methyl on C2 is on the upward bond. Because one is down and one is up, their relationship is trans. A cis relationship would require both groups to be upward or both to be downward.

PROBLEM 3-26

Name the following compounds. Remember that two *up* bonds are cis; two *down* bonds are cis; one *up* bond and one *down* bond are trans.

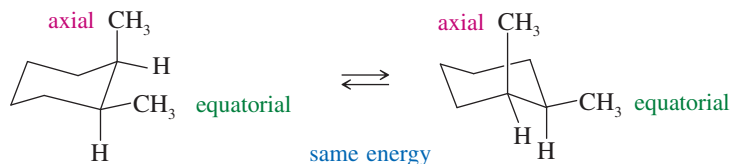


SOLVED PROBLEM 3-3

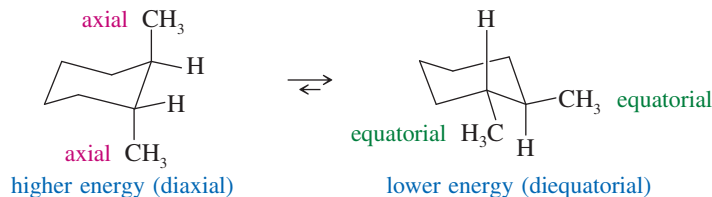
- Draw both chair conformations of *cis*-1,2-dimethylcyclohexane, and determine which conformer is more stable.
- Repeat for the *trans* isomer.
- Predict which isomer (*cis* or *trans*) is more stable.

SOLUTION

- The two possible chair conformations for the *cis* isomer interconvert at room temperature. Each of these conformations places one methyl group axial and one equatorial, giving them the same energy.



- The two chair conformations of the *trans* isomer interconvert at room temperature. Both methyl groups are axial in one, and both are equatorial in the other. The diequatorial conformation is more stable because neither methyl group occupies the more strained axial position.



- The *trans* isomer is more stable. The most stable conformation of the *trans* isomer is diequatorial and therefore about 7.6 kJ/mol (1.8 kcal/mol) lower in energy than either conformation of the *cis* isomer, each having one methyl axial and one equatorial. Remember that *cis* and *trans* are distinct isomers and cannot interconvert.

PROBLEM-SOLVING HINT

Ring-flips change the axial or equatorial positioning of groups, but they cannot change their *cis-trans* relationships. Converting *cis* into *trans* would require breaking and re-forming bonds.

PROBLEM-SOLVING HINT

If you number the carbons in a cyclohexane, the odd-numbered carbons are similar, as are the even-numbered carbons. If all the odd-numbered carbons have their up bond axial and their down bond equatorial, then all the even-numbered carbons will have their down bond axial and their up bond equatorial. For example, *cis*-1,3 (both up, both odd) will be both axial or both equatorial; *cis*-1,2 (both up, one odd, one even) will be one axial, one equatorial. This tip allows you to predict the answers before you draw them.

PROBLEM 3-27

- (a) Draw both chair conformations of *cis*-1,4-dimethylcyclohexane, and determine which conformer is more stable.
 (b) Repeat for the *trans* isomer.
 (c) Predict which isomer (*cis* or *trans*) is more stable.

PROBLEM 3-28

Use your results from Problem 3-27 to complete the following table. Each entry shows the positions of two groups arranged as shown. For example, two groups that are *trans* on adjacent carbons (*trans*-1,2) must be both equatorial (e,e) or both axial (a,a).

Positions	<i>cis</i>	<i>trans</i>
1,2	(e,a) or (a,e)	(e,e) or (a,a)
1,3		
1,4		

3-15A Substituents of Different Sizes

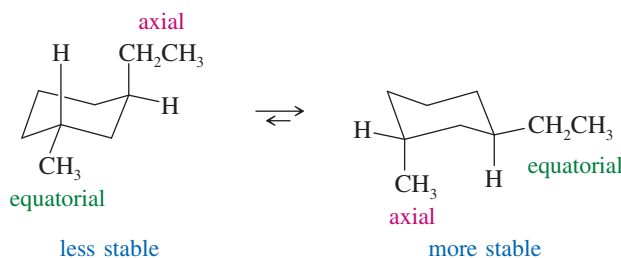
In many substituted cyclohexanes, the substituents are different sizes. As shown in Table 3-6 (page 183), the energy difference between the axial and equatorial positions for a larger group is greater than that for a smaller group. In general, if both groups cannot be equatorial, the most stable conformation has the larger group equatorial and the smaller group axial.

SOLVED PROBLEM 3-4

Draw the most stable conformation of *trans*-1-ethyl-3-methylcyclohexane.

SOLUTION

First, we draw the two conformations.



Both conformations require one group to be axial while the other is equatorial. The ethyl group is bulkier than the methyl group, so the conformation with the ethyl group equatorial is more stable. These chair conformations are in equilibrium at room temperature, and the one with the equatorial ethyl group predominates.

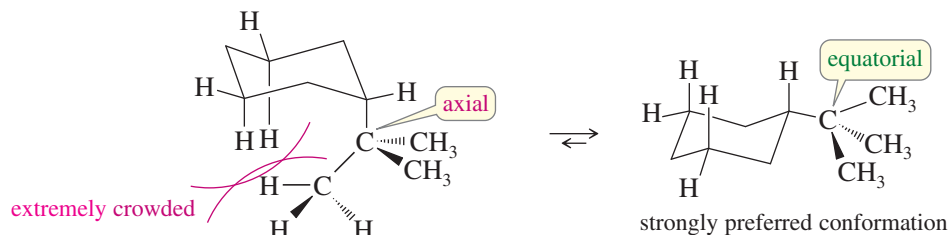
PROBLEM 3-29

Draw the two chair conformations of each of the following substituted cyclohexanes. In each case, label the more stable conformation.

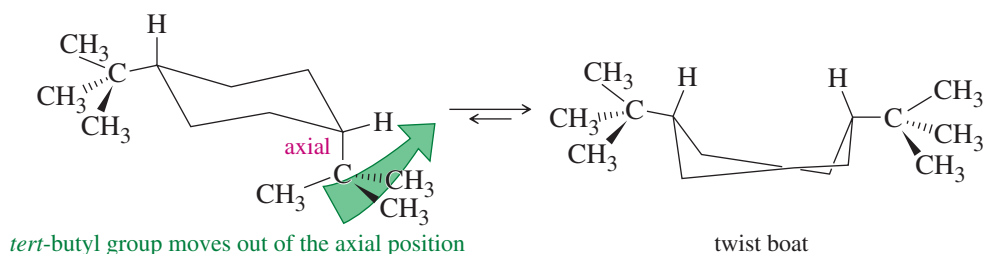
- (a) *cis*-1-ethyl-2-methylcyclohexane
 (b) *trans*-1,2-diethylcyclohexane
 (c) *cis*-1-ethyl-4-isopropylcyclohexane
 (d) *trans*-1-ethyl-4-methylcyclohexane

3-15B Extremely Bulky Groups

Some groups, such as *tertiary*-butyl groups, are so bulky that they are extremely strained in axial positions. Regardless of the other groups present, cyclohexanes with *tert*-butyl substituents are most stable when the *tert*-butyl group is in an equatorial position. The following figure shows the severe steric interactions in a chair conformation with a *tert*-butyl group axial.



If two *tert*-butyl groups are attached to the ring, both of them are much less strained in equatorial positions. When neither chair conformation allows both bulky groups to be equatorial, they may force the ring into a twist boat conformation. For example, both chair conformations of *cis*-1,4-di-*tert*-butylcyclohexane require one of the bulky *tert*-butyl groups to occupy an axial position. This compound is more stable in a twist boat conformation that allows both bulky groups to avoid axial positions.



PROBLEM 3-30

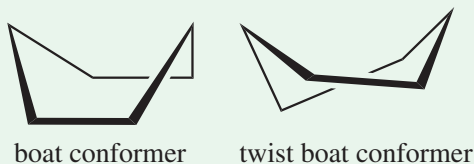
Draw the most stable conformation of

- cis*-1-*tert*-butyl-3-ethylcyclohexane.
- trans*-1-*tert*-butyl-2-methylcyclohexane.
- trans*-1-*tert*-butyl-3-(1,1-dimethylpropyl)cyclohexane.

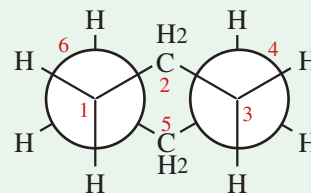
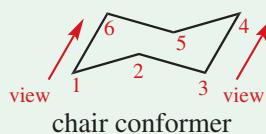
FOCUS Conformations of Cyclohexane

Rings cannot turn themselves inside out, so there is no free rotation in a cycloalkane except for very large rings. Rings are constantly in motion, however, and there are conformational extremes in cyclohexane just as in acyclic alkanes.

A high-energy conformer is the **boat** form. A boat has all eclipsed bonds, causing it to rotate into the **twist boat** (or simply twist) conformation. A boat conformer is not normally observed except in rare circumstances when constrained by bridging groups or extremely large substituents. Think of a boat form as a high-energy state between other conformers.

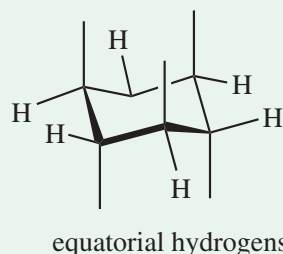
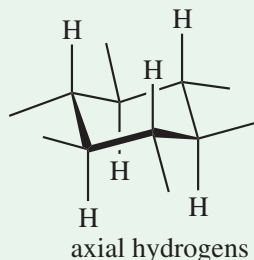


The lowest energy, most stable, and most abundant cyclohexane conformer is the **chair**.



Newman projection sighting down C1—C6 bond and C3—C4 bond

The Newman projection shows that each carbon has one bond to H that is vertical (up or down, alternating), called **axial**. The other bond to H around the equator of the ring is called **equatorial**.



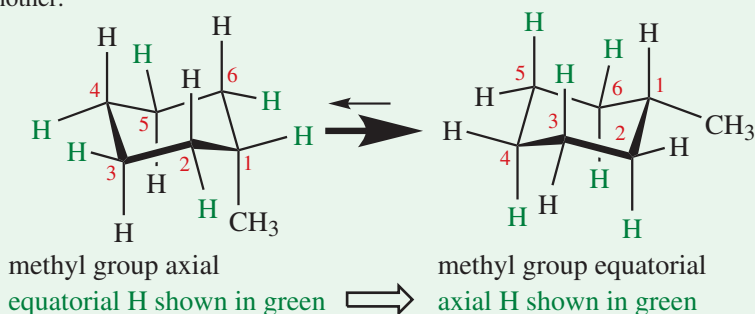
Each carbon has one axial and one equatorial bond.

Useful hint: When the bonds to the carbon point up, the axial bond at that carbon points up.

When the bonds to the carbon point down, the axial bond at that carbon points down.

Chair conformers can *flip* into one another.

During the flip, all equatorial bonds become axial, and all axial bonds become equatorial. At room temperature, unsubstituted and monosubstituted cyclohexanes flip conformations about 200,000 times per second!



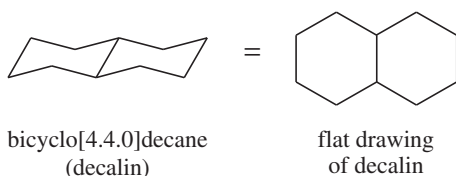
The H atoms labeled in green are the same ones in both conformers.

Conformers with substituents in equatorial positions are more stable than those with substituents in axial positions. The conformation will flip to maximize large substituents in equatorial positions.

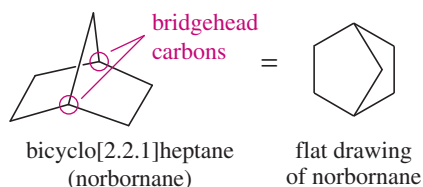
3-16 Bicyclic Molecules

Two or more rings can be joined into *bicyclic* or *polycyclic* systems. There are three ways that two rings may be joined. **Fused rings** are most common, sharing two adjacent carbon atoms and the bond between them. **Bridged rings** are also common, sharing two nonadjacent carbon atoms (the **bridgehead carbons**) and one or more carbon atoms (the bridge) between them. **Spirocyclic compounds**, in which the two rings share only one carbon atom, are relatively rare. These ring systems may be drawn realistically puckered or flat, as shown below.

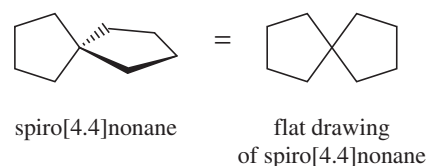
fused bicyclic



bridged bicyclic

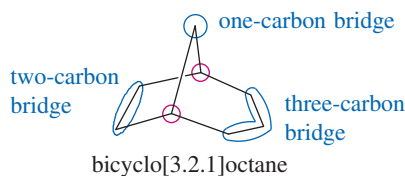
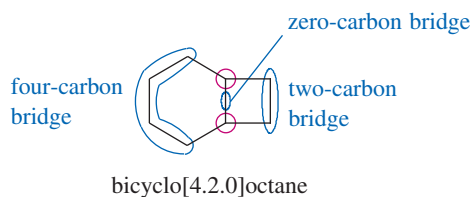


spirocyclic



3-16A Nomenclature of Bicyclic Alkanes

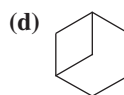
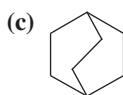
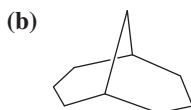
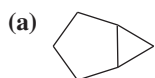
The name of a bicyclic compound is based on the name of the alkane having the same number of carbons as there are in the ring system. This name follows the prefix *bicyclo* and a set of brackets enclosing three numbers. The following examples contain eight carbon atoms and are named bicyclo[4.2.0]octane and bicyclo[3.2.1]octane, respectively.



All fused and bridged bicyclic systems have three bridges connecting the two bridgehead atoms (red circles) where the rings connect. The numbers in the brackets give the number of carbon atoms in each of the three bridges connecting the bridgehead carbons, in order of decreasing size.

PROBLEM 3-31

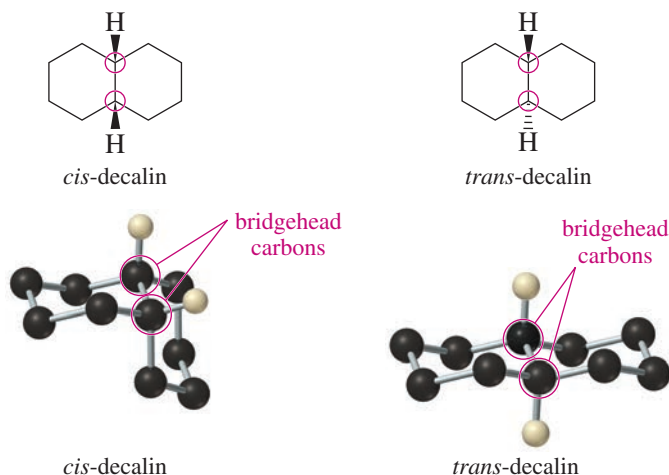
Name the following compounds.



3-16B *cis*- and *trans*-Decalin

Decalin (bicyclo[4.4.0]decane) is the most common example of a fused ring system. Two geometric isomers of decalin exist, as shown in Figure 3-27. In one isomer, the rings are fused using two *cis* bonds, while the other is fused using two *trans* bonds. You should make a model of decalin to follow this discussion.

If we consider the left ring in the drawing of *cis*-decalin, the bonds to the right ring are both directed downward (and the attached hydrogens are directed upward). These bonds are therefore *cis*, and this is a *cis* ring fusion. One of the bonds to the right ring must be axial, and the other is equatorial. In *trans*-decalin, one of the bonds to the right ring is directed upward and the other downward. These bonds are *trans*, and this is a *trans* ring fusion. Both of the bonds to the right ring are equatorial. The six-membered rings in both isomers assume chair conformations, as shown in Figure 3-27.



Application: Drugs

Bicyclic molecules are found in many natural product structures. Cocaine is a derivative of bicyclo[3.2.1]octane in which nitrogen replaces the carbon at the one-carbon bridge.

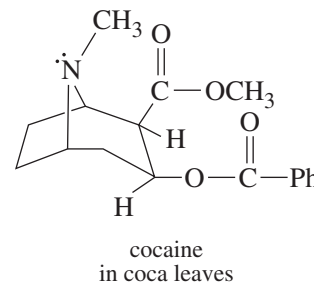


FIGURE 3-27 *cis*-Decalin has a ring fusion where the second ring is attached by two *cis* bonds. *trans*-Decalin is fused using two *trans* bonds. (The other hydrogens are omitted for clarity.)

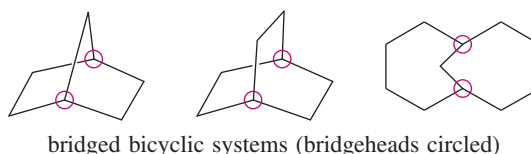
The conformation of *cis*-decalin is somewhat flexible, but the *trans* isomer is quite rigid. If one of the rings in the *trans* isomer did a chair–chair interconversion, both bonds to the second ring would become axial and would be directed 180° apart. This is an impossible conformation, and it prevents any chair–chair interconversion in *trans*-decalin.

PROBLEM 3-32

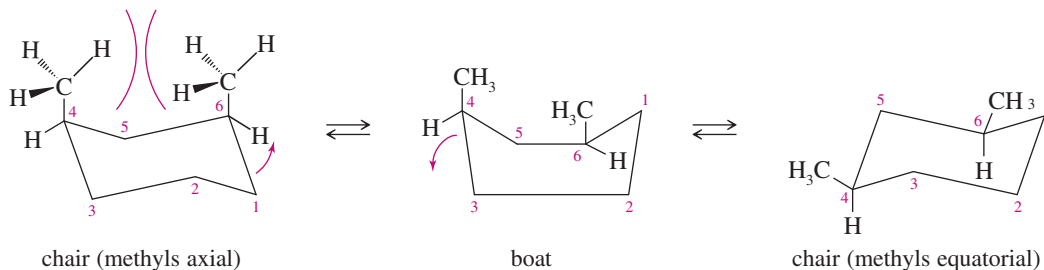
Use your models to do a chair–chair interconversion on each ring of the conformation of *cis*-decalin shown in Figure 3-27. Draw the conformation that results.

Essential Terms

acyclic	Not cyclic. (p. 170)
alkane	A hydrocarbon having only single bonds; a saturated hydrocarbon ; general formula: C_nH_{2n+2} . (p. 146)
alkyl group	The group of atoms remaining after a hydrogen atom is removed from an alkane; an alkane-like substituent. Symbolized by R. (p. 151)
amyl	An older common name for pentyl . (p. 151)
angle strain or Baeyer strain	The strain associated with distorting bond angles to smaller (or larger) angles. (p. 173)
anti conformation	A conformation with a 180° dihedral angle between the largest groups. Usually the lowest-energy conformation. (p. 167)
aromatic hydrocarbon	A hydrocarbon having a benzene-like aromatic ring. (p. 147)
axial bond	One of the six bonds (three up and three down) on the chair conformation of the cyclohexane ring that are parallel to the “axis” of the ring. The axial bonds are shown in red, and the equatorial bonds in green, in the drawing on page 178.
bridged bicyclic compound	A compound containing two rings joined at nonadjacent carbon atoms. (p. 189)

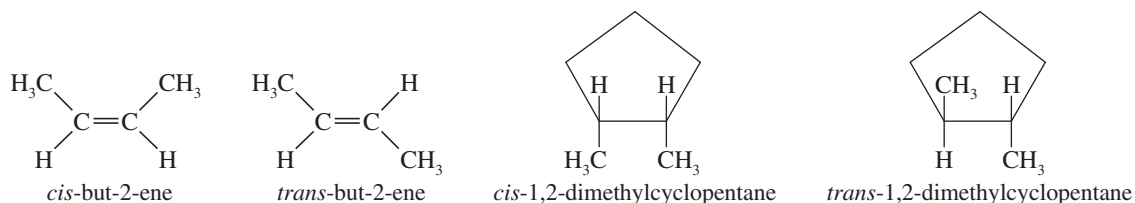


bridgehead carbons	The carbon atoms shared by two or more rings. Three chains of carbon atoms (bridges) connect the bridgeheads. (p. 188)
chair–chair interconversion	(ring-flip) The process of one chair conformation of a cyclohexane flipping into another one, with all the axial and equatorial positions reversed. The boat (or twist boat) conformation is an intermediate for the chair–chair interconversion. (p. 181)



cis-trans isomers	(geometric isomers) Stereoisomers that differ only with respect to their <i>cis</i> or <i>trans</i> arrangement on a ring or double bond. (p. 172)
cis:	Having two similar groups directed toward the same face of a ring or double bond. (p. 172)

trans: Having two similar groups directed toward opposite faces of a ring or double bond. (p. 172)



combustion

A rapid oxidation at high temperatures in the presence of air or oxygen. (p. 161)

common names

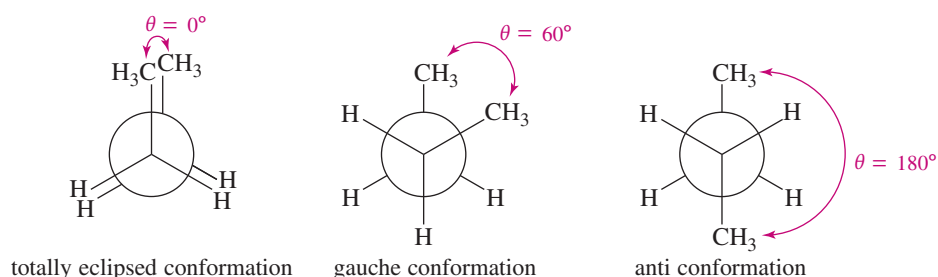
The names that have developed historically, generally with a specific name for each compound; also called **trivial names**. (p. 149)

conformational analysis

The study of the energetics of different conformations. (p. 165)

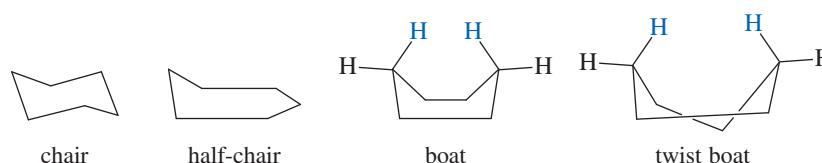
conformations and conformers

Structures that are related by rotations about single bonds. Strictly speaking, a **conformer** is a conformation that corresponds to a relative minimum in energy, usually a staggered conformation. In most cases, conformations and conformers interconvert at room temperature, and they are not true isomers. (p. 163)



conformations of cyclohexanes

(pp. 177–188)



chair conformation:

The most stable conformation of cyclohexane, with one part puckered upward and another part puckered downward. (p. 177)

boat conformation:

The less stable puckered conformation of cyclohexane, with both parts puckered upward. The most stable boat is actually the **twist boat** (or simply **twist**) conformation. Twisting minimizes torsional strain and steric strain. (p. 177)

flagpole hydrogens:

Two hydrogens (blue) in the boat conformation point upward like flagpoles. The twist boat reduces the steric repulsion of the flagpole hydrogens. (p. 177)

half-chair conformation:

The unstable conformation halfway between the chair conformation and the boat conformation. Part of the ring is flat in the half-chair conformation. (p. 178)

constitutional isomers

(**structural isomers**) Isomers whose atoms are connected differently; they differ in their bonding sequence. (p. 81)

cracking

Heating large alkanes to cleave them into smaller molecules. (p. 161)

catalytic cracking:

Cracking in the presence of a catalyst. (p. 160)

hydrocracking:

Catalytic cracking in the presence of hydrogen to give mixtures of alkanes. (p. 160)

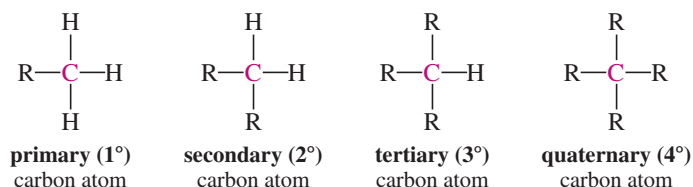
cyclic

Containing a ring of atoms. (p. 170)

cycloalkane

An alkane containing a ring of carbon atoms; general formula: C_nH_{2n} . (p. 170)

degree of alkyl substitution The number of alkyl groups bonded to a carbon atom in a compound or in an alkyl group. (p. 152)

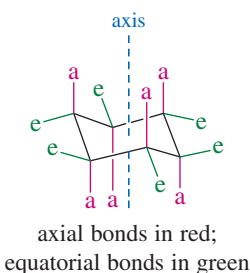


1,3-diaxial interaction The strong steric strain between two axial groups on cyclohexane carbons with one carbon between them. (p. 183)

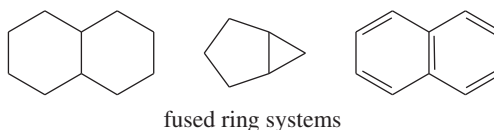
dihedral angle (θ) (see also **conformations**) The angle between two specified groups in a Newman projection. (p. 165)

eclipsed conformation Any conformation with bonds directly lined up with each other, one behind the other, in the Newman projection. The conformation with $\theta = 0^\circ$ is an eclipsed conformation. See also **staggered conformation**. (p. 165)

equatorial bond One of the six bonds (three down and three up) on the cyclohexane ring that are directed out toward the "equator" of the ring. The equatorial bonds are shown in green in the drawing below. (p. 178)



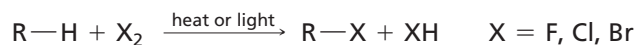
fused ring system A molecule in which two or more rings share two adjacent carbon atoms. (p. 188)



gauche conformation A conformation with a 60° dihedral angle between the largest groups. (p. 167)

geometric isomers See **cis-trans isomers**, the IUPAC term. (p. 172)

halogenation The reaction of alkanes with halogens, in the presence of heat or light, to give products with halogen atoms substituted for hydrogen atoms. (p. 162)



heat of combustion The heat given off when a mole of a compound is burned with excess oxygen to give CO_2 and H_2O in a *bomb calorimeter*. A measure of the energy content of a molecule. (p. 174)

homologs Two compounds that differ only by one or more $-\text{CH}_2-$ groups. (p. 147)

hydrophilic Attracted to water; soluble in water.

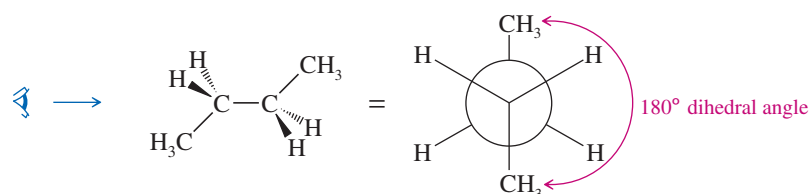
hydrophobic Repelled by water; insoluble in water. (p. 157)

IUPAC names The systematic names that follow the rules adopted by the International Union of Pure and Applied Chemistry. (p. 149)

kerosene A thin, volatile oil distilled from petroleum, with a boiling range higher than that of gasoline and lower than that of diesel fuel. Kerosene was once used in lanterns and heaters, but now most of this petroleum fraction is further refined for use as jet fuel. (p. 159)

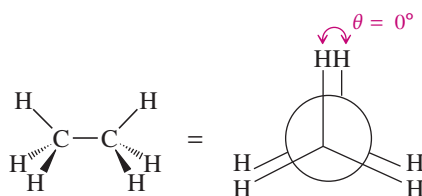
methane hydrate An ice-like substance consisting of individual methane molecules trapped inside cages of water molecules. (p. 161)

methine group	The —CH— group.
methylene group	The $\text{—CH}_2\text{—}$ group. (p. 147)
methyl group	The —CH_3 group. (p. 153)
<i>n</i>-alkane, normal alkane, or straight-chain alkane	An alkane with all its carbon atoms in a single chain, with no branching or alkyl substituents. (p. 147)
Newman projections	A way of drawing the conformations of a molecule by looking straight down the bond connecting two carbon atoms. (p. 163)

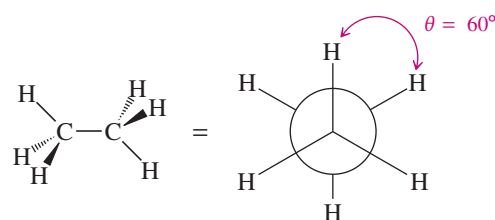


a Newman projection of butane in the anti conformation

octane number	A rating of the antiknock properties of a gasoline blend. Its octane number is the percentage of isooctane (2,2,4-trimethylpentane) in an isooctane/heptane blend that begins to knock at the same compression ratio as the gasoline being tested. (p. 159)
paraffins	Another term for alkanes. (p. 161)
ring strain	The extra strain associated with the cyclic structure of a compound, as compared with a similar acyclic compound; composed of angle strain and torsional strain. (p. 173)
angle strain or Baeyer strain:	The strain associated with distorting bond angles to smaller (or larger) angles.
torsional strain:	The strain associated with eclipsing of bonds in the ring.
saturated	Having no double or triple bonds. (p. 147)
sawhorse structures	A way of picturing conformations by looking down at an angle toward the carbon–carbon bond. (p. 165)
skew conformation	Any conformation that is not precisely staggered or eclipsed. (p. 165)
spirocyclic compounds	Bicyclic compounds in which the two rings share only one carbon atom. (p. 188)
staggered conformation	Any conformation with the bonds equally spaced in the Newman projection. The conformation with $\theta = 60^\circ$ is a staggered conformation. (p. 165)



eclipsed conformation of ethane



staggered conformation of ethane

steric strain	The interference between two bulky groups that are so close together that their electron clouds experience a repulsion. (p. 168)
substituent	A side chain or an appendage on the main chain. (p. 150)
systematic names	Same as IUPAC names, the names that follow the rules adopted by the International Union of Pure and Applied Chemistry. (p. 149)
torsional energy or conformational energy	The energy required to twist a bond into a specific conformation. (p. 165)
torsional strain	The resistance to twisting about a bond. (p. 165)
totally eclipsed conformation	A conformation with a 0° dihedral angle between the largest groups. Usually the highest-energy conformation. (p. 167)

Essential Problem-Solving Skills in Chapter 3

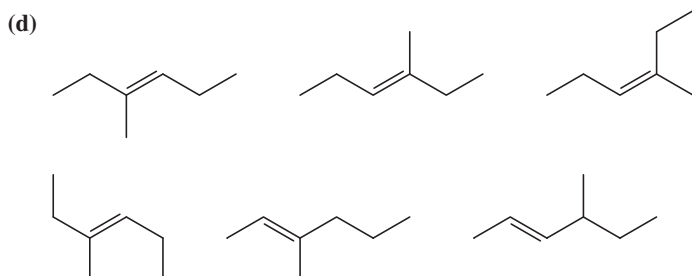
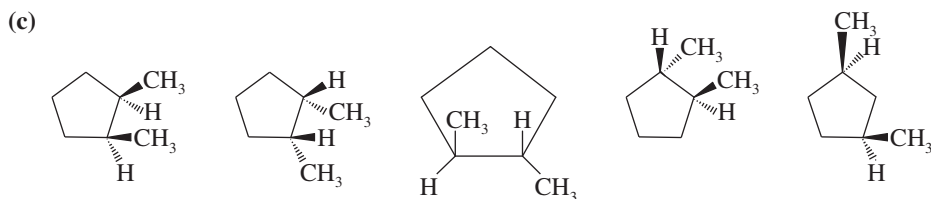
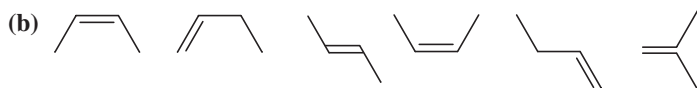
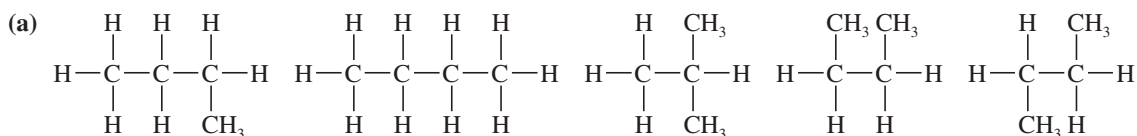
Each skill is followed by problem numbers exemplifying that particular skill.

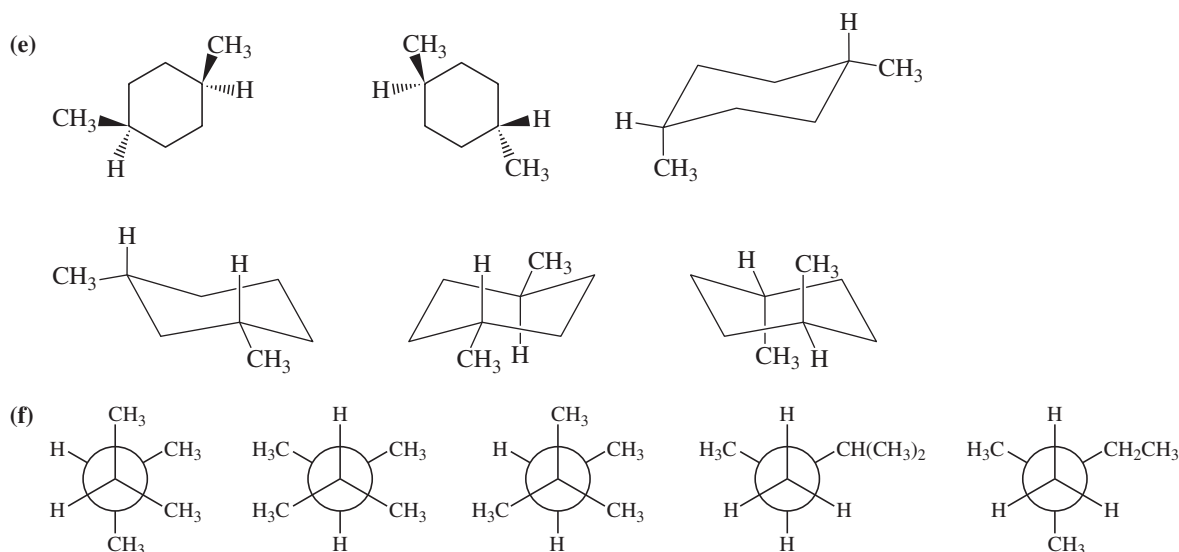
1	Given the IUPAC name or common name of an alkane, draw the structure and give the molecular formula.	Problems 3-36, 37, 41, and 49
2	Draw isomers of alkanes and cycloalkanes, and use the IUPAC rules to name alkanes, cycloalkanes, and bicyclic alkanes.	Problems 3-33, 34, 35, 39, 43, and 52
3	Explain and predict trends in the physical properties of alkanes.	Problem 3-42
4	Use Newman projections to compare conformational energies and predict the most stable conformation. Show how the torsional energy varies as the dihedral angle changes.	Problems 3-40, 44, 48, and 49
5	Compare the energies of cycloalkanes, and explain how their angle strain and torsional strain combine to give the total ring strain.	Problem 3-45
6	Draw accurate cyclohexane chair conformations. Place axial and equatorial groups in their proper positions with correct bond angles.	Problems 3-45, 46, 47, 51, and 52
	<i>Problem-Solving Strategy: Drawing Chair Conformations</i>	Problems 3-45, 46, 47, 51, and 52
7	Predict the most stable conformations of mono and disubstituted cyclohexanes by placing the larger groups in equatorial rather than axial position.	Problems 3-45, 46, 47, and 51
	<i>Problem-Solving Strategy: Recognizing Cis and Trans Isomers</i>	Problems 3-45, 46, 47, 51, and 52

Study Problems

- 3-33** (a) There are 18 isomeric alkanes of molecular formula C_8H_{18} . Draw and name any eight of them.
 (b) Draw and name the six isomeric cyclopentanes of molecular formula C_7H_{14} . These will include four constitutional isomers, of which two show geometric (cis-trans) stereoisomerism.

- 3-34** Which of the following structures represent the same compound? Which ones represent different compounds?





(g) Name the structures given in Problem 3-34, parts (a), (c), (e), and (f). Make sure your names are the same for structures that are the same, and different for structures that are different.

***3-35** (a) Draw and name the five cycloalkane structures of formula C_5H_{10} . Can any of these structures give rise to geometric (cis-trans) isomerism? If so, show the cis and trans stereoisomers.

(b) Draw and name the eight cycloalkane structures of formula C_6H_{12} that do not show geometric isomerism.

(c) Draw and name the four cycloalkanes of formula C_6H_{12} that do have cis-trans isomers.

3-36 Draw the structure that corresponds with each name.

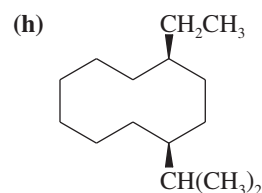
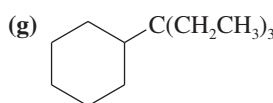
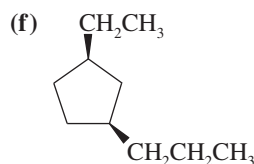
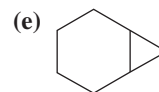
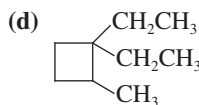
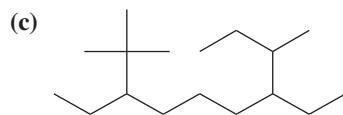
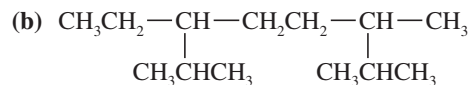
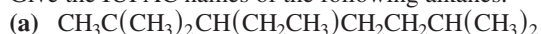
- | | | |
|---|-------------------------------|---|
| (a) 3-ethyloctane | (b) 4-isopropyldecane | (c) <i>sec</i> -butylcycloheptane |
| (d) 2,3-dimethyl-4-propylnonane | (e) 2,2,4,4-tetramethylhexane | (f) <i>trans</i> -1,3-diethylcyclopentane |
| (g) <i>cis</i> -1-ethyl-4-methylcyclohexane | (h) isobutylcyclopentane | (i) <i>tert</i> -butylcyclohexane |
| (j) pentylcyclohexane | (k) cyclobutylcyclohexane | (l) <i>cis</i> -1-bromo-3-chlorocyclohexane |

3-37 Each of the following descriptions applies to more than one alkane. In each case, draw and name two structures that match the description.

- | | | |
|--|---------------------------------------|--------------------------------------|
| (a) an isopropylheptane | (b) a diethyldecane | (c) a <i>cis</i> -diethylcyclohexane |
| (d) a <i>trans</i> -dihalocyclopentane | (e) a (2,3-dimethylpentyl)cycloalkane | (f) a bicyclononane |

3-38 Write structures for a homologous series of alkyl chlorides ($R-Cl$) having from one to six carbons.

3-39 Give the IUPAC names of the following alkanes.

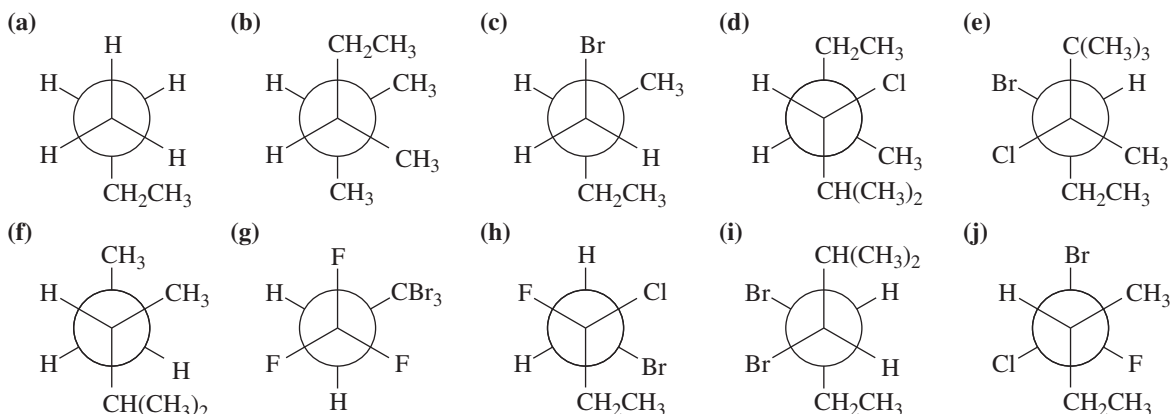


3-40 Construct a graph, similar to Figure 3-11, of the torsional energy of 3-methylpentane along the $C2-C3$ bond. Place $C2$ in front, represented by three bonds coming together in a Y shape, and $C3$ in back, represented by a circle with three bonds pointing out from it. Define the dihedral angle as the angle between the methyl group on the front carbon and the ethyl group on the back carbon. Begin your graph at the 0° dihedral angle and begin to turn the front carbon. Show the Newman projection and the approximate energy at each 60° of rotation. Indicate which conformations are the most stable (lowest energy) and the least stable (highest energy).

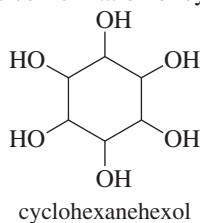
3-41 The following names are all incorrect or incomplete, but they represent real structures. Draw each structure and name it correctly.

- | | | |
|----------------------|------------------------|-----------------------------|
| (a) 2-ethylpentane | (b) 3-isopropylhexane | (c) 5-chloro-4-methylhexane |
| (d) 2-dimethylbutane | (e) 2-cyclohexylbutane | (f) 2,3-diethylcyclopentane |

- 3-42** In each pair of compounds, which compound has the higher boiling point? Explain your reasoning.
 (a) nonane or 3-methylheptane (b) octane or 2,3,4-trimethylpentane (c) 2,3,5-trimethylhexane or nonane
- 3-43** There are eight different five-carbon alkyl groups.
 (a) Draw them. (b) Give them systematic names.
 (c) In each case, label the degree of substitution (primary, secondary, or tertiary) of the head carbon atom bonded to the main chain.
- 3-44** Use a Newman projection about the indicated bond to draw the most stable conformer for each compound.
 (a) 3-methylpentane about the C2—C3 bond (b) 3,3-dimethylhexane about the C3—C4 bond
- 3-45** (a) Draw the two chair conformations of *cis*-1,3-dimethylcyclohexane, and label all the positions as axial or equatorial.
 (b) Label the higher-energy conformation and the lower-energy conformation.
 (c) The energy difference in these two conformations has been measured to be about 23 kJ (5.4 kcal) per mole. How much of this energy difference is due to the torsional energy of gauche relationships?
 (d) How much energy is due to the additional steric strain of the 1,3-diaxial interaction?
- 3-46** Draw the two chair conformations of each compound, and label the substituents as axial and equatorial. In each case, determine which conformation is more stable.
 (a) *cis*-1-ethyl-2-isopropylcyclohexane (b) *trans*-1-ethyl-2-isopropylcyclohexane
 (c) *cis*-1-ethyl-3-methylcyclohexane (d) *trans*-1-ethyl-3-methylcyclohexane
 (e) *cis*-1-ethyl-4-methylcyclohexane (f) *trans*-1-ethyl-4-methylcyclohexane
- 3-47** Using what you know about the conformational energetics of substituted cyclohexanes, predict which of the two 9-methyldecalin isomers is more stable. Explain your reasoning.
- 3-48** Convert each Newman projection to the equivalent line-angle formula, and assign the IUPAC name.



- *3-49** Draw Newman projections along the C3—C4 bond to show the most stable and least stable conformations of 3-ethyl-2,4-dimethylheptane.
- *3-50** Conformational studies on ethane-1,2-diol ($\text{HOCH}_2\text{—CH}_2\text{OH}$) have shown the most stable conformation about the central C—C bond to be the gauche conformation, which is 9.6 kJ/mol (2.3 kcal/mol) more stable than the anti conformation. Draw Newman projections of these conformers, and explain this curious result.
- 3-51** The most stable form of 1,2,3,4,5,6-cyclohexanhexol contains a six-membered ring in the chair conformation with all the substituents equatorial. Draw this most stable conformation of cyclohexanhexol.



- 3-52** This is a Newman projection of a substituted cyclohexane.
 (a) Draw the equivalent chair form.
 (b) Draw the equivalent structure using wedge and dash notation on a cyclohexane hexagon.
 (c) Give the IUPAC name.

